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AEROSOL-GENERATING SYSTEM WITH PLURALITY OF AEROSOL-GENERATING SEGMENTS

Abstract

An aerosol-generating system is provided, including: an aerosol-generating device for an aerosol-generating article, the device including: a heating chamber configured to receive the article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone; and the article to be received in the heating chamber, the article including: a first aerosol-generating segment including a first aerosol-generating substrate, and a second aerosol-generating segment including a second aerosol-generating substrate, the first substrate including between 5 and 30% by weight of glycerol on a dry weight basis, the system being configured such that when the article is fully received within the heating chamber: at least 90 percent of a length of the second segment is within the heating zone, the first segment being located upstream of the second segment, and a length of the first segment being less than the length of the second segment.

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Background/Summary

[0001] The present invention relates to an aerosol-generating system comprising an aerosol-generating device and an aerosol-generating article. In particular, the present invention relates to an aerosol-generating system in which the aerosol-generating device comprises a first aerosol-generating segment and a second aerosol-generating segment.

[0002] Aerosol-generating systems comprising an aerosol-generating device and a corresponding aerosol-generating article are known in the art. For example, systems are known in which an aerosol-generating article is heated by an aerosol-generating device to generate an aerosol. The aerosol-generating article may comprise an aerosol-generating substrate, such as a tobacco-containing substrate, which is heated rather than combusted. Typically, in such systems an aerosol is generated by the transfer of heat from a heat source of an aerosol-generating device to a physically separate aerosol-generating substrate or material which is part of an aerosol-generating article. In use, the aerosol-generating substrate may be located in contact with, within, around, or downstream of the heat source of the aerosol-generating device. During use of the aerosol-generating system, volatile compounds are released from the aerosol-generating substrate by heat transfer from the heat source and are entrained in air drawn through the aerosol-generating article. As the released compounds cool, they condense to form an aerosol.

[0003] A number of prior art documents disclose aerosol-generating devices for consuming aerosol-generating articles. Such devices include, for example, electrically heated aerosol-generating devices in which an aerosol is generated by the transfer of heat from one or more electrical heater elements of the aerosol-generating device to the aerosol-generating substrate of a heated aerosol-generating article. For example, electrically heated aerosol-generating devices have been proposed that comprise an internal heater blade which is adapted to be inserted into the aerosol-generating substrate.

[0004] It is known to provide aerosol-generating systems in which an aerosol-generating article is configured to be used with a specific aerosol-generating device.

[0005] For example, aerosol-generating articles of the prior art may include an upstream element upstream of the aerosol-generating substrate. This upstream element may be present to prevent aerosol-generating substrate from falling out of the aerosol-generating article and to control the resistance to draw of the article. Since the aerosol-generating substrate does not extend to the upstream end of the aerosol-generating article, there is no need to heat the full length of the aerosol-generating article. Accordingly, the corresponding aerosol-generating device may include a heater which, when the aerosol-generating article is fully received within the heating chamber, is located and sized to only heat the aerosol-generating substrate of the aerosol-generating article. For example, where the aerosol-generating device includes a heating chamber, the heater may not extend to the upstream end of the heating chamber.

[0006] However, the provision of an aerosol-generating device which is configured to be used with only a specific corresponding aerosol-generating article may mean the device is less effective when used with other aerosol-generating articles. For example, there may be a need to use the aerosol-generating device with an aerosol-generating article with a larger aerosol-generating substrate.

[0007] Accordingly, there is a need for an aerosol-generating system in which an aerosol-

generating device may be effectively used with alternative aerosol-generating articles to generate an aerosol. In particular, there is a need for an aerosol-generating system comprising an aerosol-generating device and an aerosol-generating article where the aerosol-generating device is suitable for heating the aerosol-generating substrate of the aerosol-generating article to a temperature sufficient for the generation of an aerosol.

[0008] According to an aspect of the present disclosure, there is provided an aerosol-generating system. The aerosol generating system may comprise an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device may comprise a heating chamber for receiving an aerosol-generating article. The aerosol-generating device may comprise a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol generating system may further comprise an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article may comprise a first aerosol-generating segment comprising a first aerosol-generating substrate. The aerosol-generating article may comprise a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system may be configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone. The aerosol-generating system may be configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The first aerosol-generating segment may be located upstream of the second aerosol-generating segment. The length of the first aerosol-generating segment may be less than the length of the second aerosol-generating segment.

[0009] According to the present invention, there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, at least 90 percent of the length of the second aerosol-generating segment is within the heating zone; wherein the first aerosol-generating segment is located upstream of the second aerosol-generating segment, and the length of the first aerosol-generating segment is less than the length of the second aerosol-generating segment.

[0010] In other words, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, most of the second aerosol-generating segment is located within the heating zone, close to the heater assembly. This may mean that the second aerosol-generating segment receives efficient, direct heating from the heater assembly.

[0011] By contrast, most of the first aerosol-generating segment may be located outside of the heating zone, more distant from the heater assembly. This may mean that the first aerosol-generating segment receives less efficient, indirect heating from the heater assembly. For example, the first aerosol-generating segment may be heated by conduction by heat from the second aerosol-generating segment.

[0012] In use, this may mean that the second aerosol-generating segment is heated to a higher temperature than the first aerosol-generating segment. This may also mean that, when the heater assembly is activated, the temperature of the second aerosol-generating segment increases faster than the temperature of the first aerosol-generating segment.

[0013] Accordingly, the provision of a first aerosol-generating segment comprising a first aerosol-

generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate, may advantageously allow the aerosol-generating substrate in each aerosol-generating segment to be configured to generate an aerosol at the temperature reached by the respective aerosol-generating segments during use. For example, the first aerosol-generating substrate may be configured to generate an aerosol at a lower temperature than the second aerosol-generating substrate.

[0014] This may advantageously allow for the total length of the aerosol-generating substrate to be greater than the length of the heating zone without compromising the efficiency of the aerosol generation since the total length of the aerosol-generating substrate may be divided into a portion which is configured to generate an aerosol at a lower temperature and a portion which is configured to generate an aerosol at a higher temperature. This arrangement may also allow for a greater total amount of aerosol-generating substrate in an aerosol-generating article to be used with an aerosol-generating device in which the heating zone is shorter than the required total length of the aerosol-generating substrate.

[0015] In a preferred embodiment there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The first aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the first aerosol-generating substrate is less than 30 percent by weight; and a second aerosol-generating segment at a location upstream of the first aerosol-generating segment and comprising a second aerosol-generating substrate, wherein the second aerosol-generating substrate comprises at least one aerosol former, wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight, on a dry weight basis.

[0016] In a preferred embodiment there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The density of the second aerosol-generating substrate is at least 1.2 times the density of the first aerosol-generating substrate.

[0017] In a preferred embodiment there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of

the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The first aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the first aerosol-generating substrate is less than 30 percent by weight; and a second aerosol-generating segment at a location upstream of the first aerosol-generating segment and comprising a second aerosol-generating substrate, wherein the second aerosol-generating substrate comprises at least one aerosol former, wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight, on a dry weight basis. The density of the second aerosol-generating substrate is at least 1.2 times the density of the first aerosol-generating substrate.

[0018] In a preferred embodiment there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The aerosol former content of the first aerosol-generating substrate is less than 30 percent by weight and the first aerosol-generating substrate has a density of less than 400 mg per cubic centimetre. The second aerosol-generating substrate comprises at least one aerosol former, wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight, on a dry weight basis and wherein the density of the second aerosol-generating substrate is greater than 500 mg per cubic centimetre.

[0019] In a preferred embodiment there is provided an aerosol-generating system comprising an aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The first aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the first aerosol-generating substrate is less than 30 percent by weight. The second aerosol-generating substrate, comprises at least one aerosol former, wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight, on a dry weight basis and wherein the ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment is no more than 0.5.

[0020] In a preferred embodiment there is provided an aerosol-generating system comprising an

aerosol-generating device for use with an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone. The aerosol-generating device further comprises an aerosol-generating article for being received in the heating chamber of the aerosol-generating device. The aerosol-generating article comprises a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate. The aerosol-generating system is configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone. The first aerosol-generating segment has a length of about 5 millimetres. The second aerosol-generating segment has a length of about 12 millimetres.

[0021] As used herein with reference to the present invention, the term “heating zone” refers to the portion of the heating chamber extending in a longitudinal direction between the upstream end of the heater assembly and the downstream end of the heater assembly.

[0022] As used herein with reference to the present invention, the term “heater assembly” refers to the component of the aerosol-generating device which is responsible for heating of the aerosol-generating substrates of the aerosol-generating article. As set out in more detail below, the heater assembly may heat the aerosol-generating substrates directly, this may be the case where the heater assembly comprises a resistive heater. The heater assembly may heat the aerosol-generating substrates indirectly, this may be the case where the heater assembly comprises an inductive coil.

[0023] As used herein with reference to the present invention, the term “longitudinal” refers to the direction corresponding to the main axis of the aerosol-generating article or aerosol-generating device, which extends between the upstream and downstream ends of the aerosol-generating article or aerosol-generating device.

[0024] As used herein with reference to the present invention, the terms “upstream” and “downstream” describe the relative positions of elements, or portions of elements, of the aerosol-generating system in relation to the direction in which the aerosol is transported through the aerosol-generating article during use. During use, air is drawn through the aerosol-generating article in the longitudinal direction.

[0025] As used herein with reference to the present invention, the term “length” denotes the dimension of a component of the aerosol-generating system in the longitudinal direction, from the component's furthest upstream point to the component's furthest downstream point. For example, it may be used to denote the dimension of aerosol-generating substrate or of any elongate tubular elements in the longitudinal direction.

[0026] The heating chamber of the aerosol-generating device may include an open downstream end and a closed upstream end. In use, the upstream end of the aerosol-generating device may be inserted into the open downstream end of the heating chamber. In use the upstream end of the aerosol-generating article may abut the upstream end of the heating chamber. Alternatively, the upstream end of the aerosol-generating article may abut another component within the heating chamber to prevent the aerosol-generating article moving any further upstream.

[0027] As used herein with reference to the present invention, the term “fully received” refers to the position when the aerosol-generating article is inserted into the heating chamber to the greatest extent it can. This may be when the upstream end of the aerosol-generating article abuts the upstream end of the heating chamber. Alternatively, this may be when the upstream end of the aerosol-generating article abuts another components within the heating chamber to prevent the aerosol-generating article moving any further upstream. When the aerosol-generating article is “fully received” in the heating chamber, a portion of the aerosol-generating article may protrude out of the open downstream end of the aerosol-generating article. This may be the case where, for example, the length of the aerosol-generating article is greater than that of the heating chamber, or

when length of the aerosol-generating article is greater than the distance between the downstream end of the heating chamber and the component within the heating chamber to prevent the aerosol-generating article moving any further upstream, where present.

[0028] When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, no more than 50 percent of the length of the first aerosol-generating segment may be within the heating zone.

[0029] For example, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device no more than 50 percent, no more than 40 percent, no more than 30 percent, no more than 20 percent, no more than 10 percent, or no more than 5 percent of the length of the first aerosol-generating segment may be within the heating zone. In some embodiments, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, none of the first aerosol-generating segment is disposed within the heating zone. In other words, the entire length of the first aerosol-generating segment may be disposed outside the heating zone.

[0030] When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the upstream end of the heating zone may be aligned with the downstream end of the first aerosol-generating segment.

[0031] When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, at least 5 percent of the length of the first aerosol-generating segment is within the heating zone.

[0032] For example, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device at least 10 percent, at least 20 percent, at least 30 percent, or at least 40 percent of the length of the first aerosol-generating segment may be within the heating zone.

[0033] When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device at least 90 percent of the length of the second aerosol-generating segment may be within the heating zone.

[0034] For example, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device at least 95 percent of the length of the second aerosol-generating segment may be within the heating zone. In some embodiments, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the entire length of the second aerosol-generating segment may be disposed within the heating zone. Where this is the case, the length of the heating zone is the same as or greater than the length of the second aerosol-generating segment.

[0035] When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the upstream end of the heating zone may be aligned with the upstream end of the second aerosol-generating segment. When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the downstream end of the heating zone may be aligned with the downstream end of the second aerosol-generating segment.

[0036] In some embodiments, when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the upstream end of the second aerosol-generating segment is aligned with the upstream end of the heating zone, the downstream end of the second aerosol-generating segment is aligned with the downstream end of the heating zone, with the first aerosol-generating segment not being disposed within the heating zone at all.

[0037] The first aerosol-generating segment may be located downstream of the second aerosol-generating segment. The first aerosol-generating segment may be located upstream of the second aerosol-generating segment. This is likely to be the case where the aerosol-generating article of the present invention is being used with an aerosol-generating device which is designed for use with a different aerosol-generating article which includes an upstream element. Where this is the case, the first aerosol-generating segment may replace all or part of the upstream element in the aerosol-

generating article.

[0038] As described in more detail below, the aerosol-generating article may further comprise an upstream element.

[0039] Alternatively, the upstream end of the first aerosol-generating segment may define the upstream end of the aerosol-generating article.

[0040] Where this is the case, the aerosol-generating article does not include an upstream element as described above. This may be the case where the aerosol-generating article of the present invention is being used with an aerosol-generating device which is designed for use with a different aerosol-generating article which includes an upstream element. Where this is the case, the first aerosol-generating segment may replace the upstream element in the aerosol-generating article.

[0041] The upstream end of the second aerosol-generating segment may be in direct contact with the downstream end of the first aerosol-generating segment. Substantially the whole surface of the downstream end of the first aerosol-generating segment may abut the upstream end of the second aerosol-generating segment.

[0042] This may advantageously improve heat transfer from the second aerosol-generating segment, which is mostly disposed within the heating zone, to the first aerosol-generating segment, which is mostly disposed outside of the heating zone.

[0043] Alternatively, the Upstream End of the Second Aerosol-Generating Segment May be Separated from the Downstream End of the First Aerosol-Generating Segment.

[0044] The length of the first aerosol-generating segment may be less than the length of the second aerosol-generating segment.

[0045] This may advantageously mean that a greater proportion of the total aerosol-generating substrate is disposed within the heating zone during use. This may advantageously improve the generation of aerosol by the aerosol-generating system.

[0046] The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be no more than 1. The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be less than 1.

[0047] For example, the ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be no more than 0.8, no more than 0.6, or no more than 0.5.

[0048] The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be at least 0.1.

[0049] For example, the ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be at least 0.2, at least 0.3, or at least 0.4.

[0050] The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be between 0.1 and 1, between 0.2 and 0.8, between 0.3 and 0.6, or between 0.4 and 0.5.

[0051] The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be about 0.4, or about 0.42.

[0052] In other embodiments, the ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be more than 1. The ratio of the length of the first aerosol-generating segment to the length of the second aerosol-generating segment may be about 1.

[0053] The length of the second aerosol-generating segment may be at least 3 millimetres greater than the length of the first aerosol-generating segment. For example, the second aerosol-generating segment may be at least 4 millimetres greater, at least 5 millimetres greater, or at least 6 millimetres greater than the length of the first aerosol-generating segment.

[0054] The length of the second aerosol-generating segment may be no more than 12 millimetres greater than the length of the first aerosol-generating segment. For example, the second aerosol-generating segment may be no more than 10 millimetres greater, no more than 9 millimetres

greater, or no more than 8 millimetres greater than the length of the first aerosol-generating segment.

[0055] The length of the second aerosol-generating segment may be about 7 millimetres greater than the length of the first aerosol-generating segment.

[0056] The first aerosol-generating segment may have a length of at least 2 millimetres. For example, the first aerosol-generating segment may have a length of at least 3 millimetres or at least 4 millimetres.

[0057] The first aerosol-generating segment may have a length of no more than 8 millimetres. For example, the first aerosol-generating segment may have a length of no more than 7 millimetres or no more than 6 millimetres.

[0058] The first aerosol-generating segment may have a length of between 2 millimetres and 8 millimetres, between 3 millimetres and 7 millimetres, or between 4 millimetres and 6 millimetres.

[0059] The first aerosol-generating segment may have a length of about 5 millimetres.

[0060] The second aerosol-generating segment may have a length of at least 8 millimetres. For example, the second aerosol-generating segment may have a length of at least 9 millimetres or at least 10 millimetres.

[0061] The second aerosol-generating segment may have a length of no more than 16 millimetres. For example, the second aerosol-generating segment may have a length of no more than 15 millimetres or no more than 14 millimetres.

[0062] The second aerosol-generating segment may have a length of between 8 millimetres and 16 millimetres, between 9 millimetres and 15 millimetres, or between 10 millimetres and 14 millimetres.

[0063] The second aerosol-generating segment may have a length of about 12 millimetres.

[0064] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be at least 10 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be at least 12 millimetres, at least 14 millimetres, or at least 16 millimetres.

[0065] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be no more than 24 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be no more than 22 millimetres, no more than 20 millimetres, or no more than 18 millimetres.

[0066] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be between 10 millimetres and 24 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be between 12 millimetres and 22 millimetres, between 14 millimetres and 20 millimetres, or between 16 millimetres and 18 millimetres.

[0067] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be about 17 millimetres.

[0068] The first aerosol-generating segment may have a length of at least 10 millimetres. For example, the first aerosol-generating segment may have a length of at least 15 millimetres or at least 18 millimetres.

[0069] The first aerosol-generating segment may have a length of no more than 30 millimetres. For example, the first aerosol-generating segment may have a length of no more than 25 millimetres or no more than 20 millimetres.

[0070] The first aerosol-generating segment may have a length of between 10 millimetres and 30 millimetres, between 15 millimetres and 25 millimetres, or between 18 millimetres and 20 millimetres.

[0071] The first aerosol-generating segment may have a length of about 18.5 millimetres. The first aerosol-generating segment may have a length of about 17 millimetres.

[0072] The second aerosol-generating segment may have a length of at least 10 millimetres. For

example, second first aerosol-generating segment may have a length of at least 15 millimetres or at least 18 millimetres.

[0073] The second aerosol-generating segment may have a length of no more than 30 millimetres. For example, the second aerosol-generating segment may have a length of no more than 25 millimetres or no more than 20 millimetres.

[0074] The second aerosol-generating segment may have a length of between 10 millimetres and 30 millimetres, between 15 millimetres and 25 millimetres, or between 18 millimetres and 20 millimetres.

[0075] The second aerosol-generating segment may have a length of about 18.5 millimetres. The second aerosol-generating segment may have a length of about 17 millimetres.

[0076] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be at least 20 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be at least 25 millimetres, at least 30 millimetres, or at least 35 millimetres.

[0077] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be no more than 55 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be no more than 50 millimetres, no more than 45 millimetres, or no more than 40 millimetres.

[0078] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be between 20 millimetres and 55 millimetres. For example, the combined length of the first aerosol-generating segment and the second aerosol-generating segment may be between 25 millimetres and 50 millimetres, between 30 millimetres and 45 millimetres, or between 35 millimetres and 40 millimetres.

[0079] The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be about 37 millimetres. The combined length of the first aerosol-generating segment and the second aerosol-generating segment may be about 34 millimetres.

[0080] The first and second aerosol-generating segments may have any external diameter. The first and second aerosol-generating segments may have substantially the same external diameter, this common diameter may be referred to as the diameter of the aerosol-generating segments. One or both of the first and second aerosol-generating segments may have an external diameter which is approximately equal to the external diameter of the aerosol-generating article.

[0081] The “external diameter” of an aerosol-generating segment may be calculated as the average of a plurality of measurements of the diameter of the aerosol-generating segment taken at different locations along the length of the aerosol-generating segment.

[0082] Preferably, the aerosol-generating segments have an external diameter of at least about 5 millimetres. More preferably, the aerosol-generating segments have an external diameter of at least about 6 millimetres. Even more preferably, the aerosol-generating segments have an external diameter of at least about 7 millimetres.

[0083] The aerosol-generating segments preferably have an external diameter of less than or equal to about 12 millimetres. More preferably, the aerosol-generating segments have an external diameter of less than or equal to about 10 millimetres. Even more preferably, the aerosol-generating segments have an external diameter of less than or equal to about 8 millimetres.

[0084] In general, it has been observed that the smaller the diameter of the aerosol-generating segments, the lower the temperature that is required to raise a core temperature of the aerosol-generating segments such that sufficient amounts of vaporizable species are released from the aerosol-generating substrates to form a desired amount of aerosol.

[0085] The aerosol-generating device may comprise a body. The body or housing of the aerosol-generating device may define a heating chamber for removably receiving the aerosol-generating article at the downstream end of the device.

[0086] The length of the heating chamber may be between 15 millimetres and 80 millimetres.

Preferably, the length of the heating chamber is between 20 millimetres and 70 millimetres. More preferably, the length of the heating chamber is between 25 millimetres and 60 millimetres. More preferably, the length of the heating chamber is between 25 millimetres and 50 millimetres.

[0087] The length of the heating chamber may be between 25 millimetres and 29 millimetres.

Preferably, the length of the heating chamber is between 25 millimetres and 29 millimetres. More preferably, the length of the heating chamber is between 26 millimetres and 29 millimetres. Even more preferably, the length of the heating chamber is 27 millimetres or 28 millimetres.

[0088] The length of the heating chamber may be the same as or greater than the combined length of the first aerosol-generating segment and the second aerosol-generating segment. Preferably, the length of the heating chamber is such that at least 75 percent of the combined length of the first aerosol-generating segment and the second aerosol-generating segment is inserted or received within the device heating chamber, when the aerosol-generating article is fully received within the heating chamber. More preferably, the length of the heating chamber is such that at least 80 percent of the combined length of the first aerosol-generating segment and the second aerosol-generating segment is inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. More preferably, the length of the heating chamber is such that at least 90 percent of the combined length of the first aerosol-generating segment and the second aerosol-generating segment is inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. This maximises the length of the first aerosol-generating segment and second aerosol-generating segment along which the first and second aerosol-generating substrates can be heated during use, thereby optimising the generation of aerosol from the aerosol-generating substrates and reducing waste.

[0089] The length of the heating chamber may be such that the downstream section or a portion thereof is configured to protrude from the heating chamber, when the aerosol-generating article is fully received within the heating chamber. The length of the heating chamber may be such that a portion of the downstream section (such as the hollow tubular cooling element or downstream filter segment) is configured to protrude from the heating chamber, when the aerosol-generating article is fully received within the heating chamber. The length of the heating chamber may be such that a portion of the downstream section (such as the hollow tubular cooling element or downstream filter segment) is configured to be received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber.

[0090] At least 25 percent of the length of the downstream section may be inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. At least 30 percent of the length of the downstream section may be inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber.

[0091] At least 30 percent of the length of the hollow tubular element may be inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. At least 40 percent of the length of the hollow tubular element may be inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. At least 50 percent of the length of the hollow tubular element may be inserted or received within the heating chamber, when the aerosol-generating article is fully received within the heating chamber. Various lengths of the hollow tubular element are described in more detail within the present disclosure.

[0092] Optimising the amount or length of the aerosol-generating article that is inserted into the heating chamber of the aerosol-generating device may enhance the article's resistance to inadvertently falling out during use. Particularly, during the heating of the aerosol-generating substrates, the substrates may shrink such that its external diameter may have reduced, thereby reducing the extent to which the inserted portion of the article inserted into the device can frictionally engage with the heating chamber. The inserted portion of the article, or the portion of

the article configured to be received within the heating chamber, may be the same length as the heating chamber.

[0093] A diameter of the heating chamber may be between 4 millimetres and 10 millimetres. A diameter of the heating chamber may be between 5 millimetres and 9 millimetres. A diameter of the heating chamber may be between 6 millimetres and 8 millimetres. A diameter of the heating chamber may be between 6 millimetres and 7 millimetres.

[0094] A diameter of the heating chamber may be substantially the same as or greater than a diameter of the aerosol-generating article. A diameter of the heating chamber may be the same as a diameter of the aerosol-generating article in order to establish a tight fit with the aerosol-generating article.

[0095] The heating chamber may be configured to establish a tight fit with an aerosol-generating article received within the heating chamber. Tight fit may refer to a snug fit. The aerosol-generating device may comprise a peripheral wall. Such a peripheral wall may define the heating chamber. The peripheral wall defining the heating chamber may be configured to engage with an aerosol-generating article received within the heating chamber in a tight fit manner, so that there is substantially no gap or empty space between the peripheral wall defining the heating chamber and the aerosol-generating article when received within heating chamber.

[0096] Such a tight fit may establish an airtight fit or configuration between the heating chamber and an aerosol-generating article received therein.

[0097] With such an airtight configuration, there would be substantially no gap or empty space between the peripheral wall defining the heating chamber and the aerosol-generating article for air to flow through.

[0098] The tight fit with an aerosol-generating article may be established along the entire length of the heating chamber or along a portion of the length of the heating chamber.

[0099] The aerosol-generating device may comprise an air-flow channel extending between a channel inlet and a channel outlet. The air-flow channel may be configured to establish a fluid communication between the interior of the heating chamber and the exterior of the aerosol-generating device. The air-flow channel of the aerosol-generating device may be defined within the housing of the aerosol-generating device to enable fluid communication between the interior of the heating chamber and the exterior of the aerosol-generating device. When an aerosol-generating article is received within the heating chamber, the air-flow channel may be configured to provide air flow into the article in order to deliver generated aerosol to a user drawing from the downstream end of the article.

[0100] The air-flow channel of the aerosol-generating device may be defined within, or by, the peripheral wall of the housing of the aerosol-generating device. In other words, the air-flow channel of the aerosol-generating device may be defined within the thickness of the peripheral wall or by the inner surface of the peripheral wall, or a combination of both. The air-flow channel may partially be defined by the inner surface of the peripheral wall and may be partially defined within the thickness of the peripheral wall. The inner surface of the peripheral wall defines a peripheral boundary of the heating chamber.

[0101] The air-flow channel of the aerosol-generating device may extend from an inlet located at the mouth end, of the aerosol-generating device to an outlet located away from mouth end of the device. The air-flow channel may extend along a direction parallel to the longitudinal axis of the aerosol-generating device.

[0102] The heater assembly may comprise any suitable type of heating element or heater. The heater assembly may comprise at least one of a resistive heating element and an inductive heating assembly. The heater assembly may comprise an external heater or external heating element.

[0103] The heater assembly may externally heat the aerosol-generating article when received within the aerosol-generating device. Such an external heater assembly may circumscribe the aerosol-generating article when inserted in or received within the heating chamber of the aerosol-

generating device.

[0104] In some embodiments, the heater assembly is arranged to heat the outer surface of at least one of the aerosol-generating substrates. The heater assembly may be positioned within the heating chamber.

[0105] The heater assembly may comprise at least one resistive heating element. The at least one resistive heating element may be any suitable type of resistive heating element. In some embodiments, the heater assembly comprises only one resistive heating element. In some embodiments, the heater assembly comprises a plurality of resistive heating elements. The heater may comprise at least one resistive heating element. Preferably, the heater assembly comprises a plurality of resistive heating elements. Preferably, the resistive heating elements are electrically connected in a parallel arrangement. Advantageously, providing a plurality of resistive heating elements electrically connected in a parallel arrangement may facilitate the delivery of a desired electrical power to the heater while reducing or minimising the voltage required to provide the desired electrical power. Advantageously, reducing or minimising the voltage required to operate the heater may facilitate reducing or minimising the physical size of the power supply.

[0106] Suitable materials for forming the at least one resistive heating element include but are not limited to: semiconductors such as doped ceramics, electrically 'conductive' ceramics (such as, for example, molybdenum disilicide), carbon, graphite, metals, metal alloys and composite materials made of a ceramic material and a metallic material. Such composite materials may comprise doped or undoped ceramics. Examples of suitable doped ceramics include doped silicon carbides.

Examples of suitable metals include titanium, zirconium, tantalum and metals from the platinum group. Examples of suitable metal alloys include stainless steel, nickel-, cobalt-, chromium-, aluminium-titanium-zirconium-, hafnium-, niobium-, molybdenum-, tantalum-, tungsten-, tin-, gallium-, manganese- and iron-containing alloys, and super-alloys based on nickel, iron, cobalt, stainless steel, Timetal® and iron-manganese-aluminium based alloys.

[0107] In some embodiments, the at least one resistive heating element comprises one or more stamped portions of electrically resistive material, such as stainless steel. Alternatively, the at least one resistive heating element may comprise a heating wire or filament, for example a Ni—Cr (Nickel-Chromium), platinum, tungsten or alloy wire.

[0108] In some embodiments, the at least one heating element comprises an electrically insulating substrate, wherein the at least one resistive heating element is provided on the electrically insulating substrate.

[0109] The electrically insulating substrate may comprise any suitable material. For example, the electrically insulating substrate may comprise one or more of: paper, glass, ceramic, anodized metal, coated metal, and a polymer. The ceramic may comprise mica, Alumina (Al₂O₃) or Zirconia (ZrO₂). The polymer may comprise a Polyimide. Preferably, the electrically insulating substrate has a thermal conductivity of less than or equal to about 40 Watts per metre Kelvin, preferably less than or equal to about 20 Watts per metre Kelvin and ideally less than or equal to about 2 Watts per metre Kelvin.

[0110] Where the heater assembly comprises a resistive heating element, the resistive heating element may circumscribe at least a portion of the heating chamber, defining the heating zone.

[0111] In some embodiments, the heater assembly comprises an inductive heating assembly. The inductive heating assembly may comprise an inductor coil. The aerosol-generating device may comprise a power supply configured to provide high frequency oscillating current to the inductor coil. As used herein, a high frequency oscillating current means an oscillating current having a frequency of between about 500 kHz and about 30 MHz. The aerosol-generating device may advantageously comprise a DC/AC inverter for converting a DC current supplied by a DC power supply to the alternating current. The inductor coil may be arranged to generate a high frequency oscillating electromagnetic field on receiving a high frequency oscillating current from the power supply. The inductor coil may be arranged to generate a high frequency oscillating electromagnetic

field in the heating chamber. In some embodiments, the inductor coil may substantially circumscribe the heating chamber. The inductor coil may extend at least partially along the length of the heating chamber, defining the heating zone.

[0112] The heater assembly may comprise an inductively heated element. The inductively heated element may be a susceptor element. As used herein with reference to the invention, the term 'susceptor element' refers to an element comprising a material that is capable of converting electromagnetic energy into heat. When a susceptor element is located in an alternating electromagnetic field, the susceptor is heated. Heating of the susceptor element may be the result of at least one of hysteresis losses and eddy currents induced in the susceptor, depending on the electrical and magnetic properties of the susceptor material.

[0113] A susceptor element may be arranged such that, when the aerosol-generating article is received in the hearing chamber of the aerosol-generating device, the oscillating electromagnetic field generated by the inductor coil induces a current in the susceptor element, causing the susceptor element to heat up. In these embodiments, the aerosol-generating device is preferably capable of generating a fluctuating electromagnetic field having a magnetic field strength (H-field strength) of between 1 and 5 kilo amperes per metre (kA m), preferably between 2 and 3 kA/m, for example about 2.5 kA/m. The electrically-operated aerosol-generating device is preferably capable of generating a fluctuating electromagnetic field having a frequency of between 1 and 30 MHz, for example between 1 and 10 MHz, for example between 5 and 7 MHz.

[0114] In these embodiments, the susceptor element is preferably located in contact with the aerosol-generating substrates. In some embodiments, a susceptor element is located in the aerosol-generating device. In these embodiments, the susceptor element may be located in the hearing chamber. The aerosol-generating device may comprise only one susceptor element. The aerosol-generating device may comprise a plurality of susceptor elements. In some embodiments, the susceptor element is preferably arranged to heat the outer surface of the aerosol-generating substrates.

[0115] Where the heater assembly comprises both an induction coil and an inductively heated element, the heating zone is defined as the longitudinal space between the most upstream portion of the induction claim and the inductively heated element, and the most downstream portion of the induction claim and the inductively heated element.

[0116] The susceptor element may comprise any suitable material. The susceptor element may be formed from any material that can be inductively heated to a temperature sufficient to release volatile compounds from the aerosol-generating substrate. Suitable materials for the elongate susceptor element include graphite, molybdenum, silicon carbide, stainless steels, niobium, aluminium, nickel, nickel containing compounds, titanium, and composites of metallic materials. Some susceptor elements comprise a metal or carbon. Advantageously the susceptor element may comprise or consist of a ferromagnetic material, for example, ferritic iron, a ferromagnetic alloy, such as ferromagnetic steel or stainless steel, ferromagnetic particles, and ferrite. A suitable susceptor element may be, or comprise, aluminium. The susceptor element preferably comprises more than about 5 percent, preferably more than 20 percent, more preferably more than 50 percent or more than 90 percent of ferromagnetic or paramagnetic materials. Some elongate susceptor elements may be heated to a temperature in excess of 250 degrees Celsius.

[0117] The susceptor element may comprise a non-metallic core with a metal layer disposed on the non-metallic core. For example, the susceptor element may comprise metallic tracks formed on an outer surface of a ceramic core or substrate.

[0118] As described in more detail below, in some embodiments where the aerosol-generating device comprises an induction coil, the aerosol-generating article may comprise at least one susceptor element.

[0119] In some embodiments the aerosol-generating device may comprise at least one resistive heating element and at least one inductive heating assembly. In some embodiments the aerosol-

generating device may comprise a combination of resistive heating elements and inductive heating assembly.

[0120] The first aerosol-generating substrate has a first density and the second aerosol-generating substrate has a second density, the second density may be greater than the first density.

[0121] As used herein with reference to the present invention, the term 'density' refers to the bulk density of the first and second aerosol-generating substrates. This can be calculated by measuring the total weight of the aerosol-generating substrate and dividing this by the volume of the segment of aerosol-generating substrate (excluding any wrapper).

[0122] The provision of a first aerosol-generating substrate having a lower density may advantageously provide improved aerosol delivery of the aerosol-generating article. The first aerosol-generating substrate, which has a lower density, will heat up faster than it would if it had a higher density due to the lower thermal inertia. This is advantageous since the majority of the first aerosol-generating substrate is not located in the heating zone and so will be heated to a lower temperature than the second aerosol-generating substrate. Providing a lower density in the first aerosol-generating substrate may advantageously allow for effective aerosol-generation in the first aerosol-generating substrate despite the indirect heating most of the aerosol-generating substrate will receive. By contrast, the second aerosol-generating substrate which is mostly located within the heating zone, can have a higher density as the second aerosol-generating substrate is more efficiently heated. The higher density in the second aerosol-generating substrate may advantageously allow for sustained aerosol delivery over the full duration of the use of the aerosol-generating article.

[0123] Preferably, the first aerosol-generating substrate has a density of at least 100 mg per cubic centimetre. More preferably, the density of the first aerosol-generating substrate is at least 125 mg per cubic centimetre. More preferably, the density of the first aerosol-generating substrate is at least 150 mg per cubic centimetre. Even more preferably, the density of the first aerosol-generating substrate is at least 200 mg per cubic centimetre.

[0124] The provision of a first aerosol-generating substrate having a density of at least 100 mg per cubic centimetre may advantageously prevent the first aerosol-generating substrate from falling out of, or otherwise being dislodged from the upstream end of the first aerosol-generating segment where the upstream end of the first aerosol-generating segment defines the upstream end of the aerosol-generating article.

[0125] Preferably, the first aerosol-generating substrate has a density of less than 400 mg per cubic centimetre. More preferably, the density of the first aerosol-generating substrate is less than 375 mg per cubic centimetre. More preferably, the density of the first aerosol-generating substrate is less than 350 mg per cubic centimetre. More preferably, the density of the first aerosol-generating substrate is less than 300 mg per cubic centimetre.

[0126] For example, the first aerosol-generating substrate may have a density of between 100 mg per cubic centimetre and 400 mg per cubic centimetre, or between 125 mg per cubic centimetre and 375 mg per cubic centimetre, or between 150 mg per cubic centimetre and 350 mg per cubic centimetre, or between 200 mg per cubic centimetre and 300 mg per cubic centimetre.

[0127] Preferably, the second aerosol-generating substrate has a density of at least 500 mg per cubic centimetre. More preferably, the density of the second aerosol-generating substrate is at least 525 mg per cubic centimetre. More preferably, the density of the second aerosol-generating substrate is at least 550 mg per cubic centimetre. Even more preferably, the density of the second aerosol-generating substrate is at least 600 mg per cubic centimetre.

[0128] Preferably, the second aerosol-generating substrate has a density of less than 1000 mg per cubic centimetre. More preferably, the density of the second aerosol-generating substrate is less than 900 mg per cubic centimetre. More preferably, the density of the second aerosol-generating substrate is less than 800 mg per cubic centimetre. More preferably, the density of the second aerosol-generating substrate is less than 750 mg per cubic centimetre.

[0129] For example, the second aerosol-generating substrate may have a density of between 500 mg per cubic centimetre and 1000 mg per cubic centimetre, or between 525 mg per cubic centimetre and 900 mg per cubic centimetre, or between 550 mg per cubic centimetre and 880 mg per cubic centimetre, or between 600 mg per cubic centimetre and 750 mg per cubic centimetre.

[0130] Preferably, the density of the second aerosol-generating substrate is at least 25 mg per cubic centimetre higher than the density of the first aerosol-generating substrate, or at least 50 mg per cubic centimetre higher than the density of the first aerosol-generating substrate, or at least 75 mg per cubic centimetre higher than the density of the first aerosol-generating substrate, or is at least 100 mg per cubic centimetre higher than the density of the first aerosol-generating substrate. More preferably, the density of the second aerosol-generating substrate is at least 150 mg per cubic centimetre higher than the density of the first aerosol-generating substrate. More preferably, the density of the second aerosol-generating substrate is at least 200 mg per cubic centimetre higher than the density of the first aerosol-generating substrate. The density of the second aerosol-generating substrate may be up to 500 mg per cubic centimetre higher than the density of the first aerosol-generating substrate.

[0131] Preferably, the density of the second aerosol-generating substrate is at least 1.05 times the density of the first aerosol-generating substrate, or at least 1.1 times the density of the first aerosol-generating substrate, or at least 1.2 times the density of the first aerosol-generating substrate. More preferably, the density of the second aerosol-generating substrate is at least 1.3 times the density of the first aerosol-generating substrate, or at least 1.4 times the density of the first aerosol-generating substrate, or at least 1.5 times the density of the first aerosol-generating substrate. More preferably, the density of the second aerosol-generating substrate is at least twice the density of the first aerosol-generating substrate. The density of the second aerosol-generating substrate may be up to 4 times the density of the first aerosol-generating substrate.

[0132] In preferred embodiments of the invention, the density of the second aerosol-generating substrate is at least 1.2 times the density of the first aerosol-generating substrate and the aerosol former content of the second aerosol-generating substrate is at least twice the aerosol former content of the first aerosol-generating substrate.

[0133] The first aerosol-generating substrate and the second aerosol-generating substrate may be formed of the same type of substrate as each other. Suitable types of materials for use in the aerosol-generating article of the present invention are described below and include, for example, tobacco cut filler, homogenised tobacco material such as cast leaf and aerosol-generating films. Preferably, the first aerosol-generating substrate and the second aerosol-generating substrate are different types of material to each other.

[0134] Preferably, the first aerosol-generating substrate comprises tobacco material. In certain preferred embodiments, the first aerosol-generating substrate comprises shredded tobacco material. For example, the shredded tobacco material may be in the form of cut filler, as described in more detail below. Alternatively, the shredded tobacco material may be in the form of a shredded sheet of homogenised tobacco material. Suitable homogenised tobacco materials for use in the present invention are described below.

[0135] Within the context of the present specification, the term “cut filler” is used to describe to a blend of shredded plant material, such as tobacco plant material, including, in particular, one or more of leaf lamina, processed stems and ribs, homogenised plant material.

[0136] The cut filler may also comprise other after-cut, filler tobacco or casing.

[0137] Preferably, the cut filler comprises at least 25 percent of plant leaf lamina, more preferably, at least 50 percent of plant leaf lamina, still more preferably at least 75 percent of plant leaf lamina and most preferably at least 90 percent of plant leaf lamina. Preferably, the plant material is one of tobacco, mint, tea and cloves. Most preferably, the plant material is tobacco. However, the invention is equally applicable to other plant material that has the ability to release substances upon the application of heat that can subsequently form an aerosol.

[0138] Preferably, the cut filler comprises tobacco plant material comprising lamina of one or more of bright tobacco, dark tobacco, aromatic tobacco and filler tobacco. With reference to the present invention, the term “tobacco” describes any plant member of the genus *Nicotiana*.

[0139] The cut filler suitable to be used with the present invention generally may resemble cut filler used for conventional smoking articles. The cut width of the cut filler preferably is between 0.3 millimetres and 2.0 millimetres, more preferably, the cut width of the cut filler is between 0.5 millimetres and 1.2 millimetres and most preferably, the cut width of the cut filler is between 0.6 millimetres and 0.9 millimetres. The cut width may play a role in the distribution of heat inside the first aerosol-generating substrate. Also, the cut width may play a role in the resistance to draw of the article. Further, the cut width may impact the overall density of the aerosol-generating substrate as a whole.

[0140] The strand length of the cut-filler is to some extent a random value as the length of the strands will depend on the overall size of the object that the strand is cut off from. Nevertheless, by conditioning the material before cutting, for example by controlling the moisture content and the overall subtlety of the material, longer strands can be cut. Preferably, the strands have a length of between about 10 millimetres and about 40 millimetres before the strands are collated to form the first aerosol-generating substrate. Obviously, if the strands are arranged in the first aerosol-generating substrate in a longitudinal extension where the longitudinal extension of the section is below 40 millimetres, the first aerosol-generating substrate may comprise strands that are on average shorter than the initial strand length. Preferably, the strand length of the cut-filler is such that between about 20 percent and 60 percent of the strands extend along the full length of the first aerosol-generating segment. This prevents the strands from dislodging easily from the first aerosol-generating segment.

[0141] In other preferred embodiments, the first aerosol-generating substrate comprises homogenised plant material, preferably a homogenised tobacco material.

[0142] As used herein, the term “homogenised plant material” encompasses any plant material formed by the agglomeration of particles of plant. For example, sheets or webs of homogenised tobacco material for the aerosol-generating substrates of the present invention may be formed by agglomerating particles of tobacco material obtained by pulverising, grinding or comminuting plant material and optionally one or more of tobacco leaf lamina and tobacco leaf stems. The homogenised plant material may be produced by casting, extrusion, paper making processes or other any other suitable processes known in the art.

[0143] The homogenised plant material can be provided in any suitable form.

[0144] In some embodiments, the homogenised plant material may be in the form of one or more sheets. As used herein with reference to the invention, the term “sheet” describes a laminar element having a width and length substantially greater than the thickness thereof.

[0145] The second aerosol-generating substrate may be in the form of cut filler, or a homogenised tobacco material, as described above.

[0146] Preferably, the second aerosol-generating substrate is in the form of an aerosol-generating film comprising a cellulosic based film forming agent, nicotine and the aerosol former. The aerosol-generating film may further comprise a cellulose based strengthening agent. The aerosol-generating film may further comprise water, preferably 30 percent by weight or less of water.

[0147] As used herein, the term “film” is used to describe a solid laminar element having a thickness that is less than the width or length thereof. The film may be self-supporting. In other words, a film may have cohesion and mechanical properties such that the film, even if obtained by casting a film-forming formulation on a support surface, can be separated from the support surface. Alternatively, the film may be disposed on a support or sandwiched between other materials. This may enhance the mechanical stability of the film.

[0148] The aerosol former content of the aerosol-generating film is within the ranges defined for the second aerosol-generating substrate.

[0149] In the context of the present invention the term “cellulose based film-forming agent” is used to describe a cellulosic polymer capable, by itself or in the presence of an auxiliary thickening agent, of forming a continuous film.

[0150] Preferably, the cellulose based film-forming agent is selected from the group consisting of hydroxypropyl methylcellulose (HPMC), methylcellulose (MC), ethylcellulose (EC), hydroxyethyl methyl cellulose (HEMC), hydroxyethyl cellulose (HEC), hydroxypropyl cellulose (HPC), and combinations thereof.

[0151] More preferably, the cellulose based film-forming agent is selected from the group consisting of hydroxypropyl methylcellulose (HPMC), methylcellulose (MC), ethylcellulose (EC), and combinations thereof.

[0152] In particularly preferred embodiments, the cellulose based film-forming agent is HPMC.

[0153] The aerosol-generating film for forming the second aerosol-generating substrate preferably comprises at least 10 percent by weight of a cellulose based film-forming agent, on a dry weight basis. More preferably, the aerosol-generating film comprises at least 15 percent by weight of a cellulose based film-forming agent on a dry weight basis. More preferably, the aerosol-generating film comprises at least 20 percent by weight of a cellulose based film forming agent on a dry weight basis.

[0154] Preferably, the aerosol-generating film comprises no more than 40 percent by weight of a cellulose based film-forming agent on a dry weight basis. More preferably, the aerosol-generating film comprises no more than 35 percent by weight of a cellulose based film-forming agent on a dry weight basis. More preferably, the aerosol-generating film comprises no more than 30 percent weight of a cellulose based film-forming agent on a dry weight basis.

[0155] For example, the aerosol-generating film may have a cellulose based film-forming agent content of between 10 percent and 40 percent by weight, or between 15 percent and 35 percent by weight, or between 20 percent and 30 percent by weight, on a dry weight basis.

[0156] Preferably, the aerosol-generating film further comprises a cellulose based strengthening agent. Preferably, the cellulose based strengthening agent is selected from the group consisting of cellulose fibres, microcrystalline cellulose (MCC), cellulose powder, and combinations thereof.

[0157] The aerosol-generating film preferably comprises at least 0.5 percent by weight of the cellulose based strengthening agent on a dry weight basis. More preferably, the aerosol-generating film comprises at least 5 percent by weight of the cellulose based strengthening agent on a dry weight basis. More preferably, the aerosol-generating film comprises at least 10 percent by weight of the cellulose based strengthening agent on a dry weight basis.

[0158] Preferably, the aerosol-generating film comprises no more than 40 percent by weight of the cellulose based strengthening agent on a dry weight basis. More preferably, the aerosol-generating film comprises no more than 30 percent by weight of the cellulose based strengthening agent on a dry weight basis. More preferably, the aerosol-generating film comprises no more than 25 percent by weight of the cellulose based strengthening agent on a dry weight basis.

[0159] For example, the aerosol-generating film may have a cellulose based strengthening agent content of between 0.5 percent and 40 percent by weight on a dry weight basis, or between 5 percent and 30 percent by weight on a dry weight basis, or between 10 percent and 25 percent by weight on a dry weight basis.

[0160] The aerosol-generating film may further comprise a carboxymethyl cellulose, preferably sodium carboxymethyl cellulose.

[0161] The aerosol-generating film may have a carboxymethyl cellulose content of greater than about 1 percent by weight on a dry weight basis. The aerosol-generating film may have a carboxymethyl cellulose content of greater than about 2 percent by weight on a dry weight basis. The aerosol-generating film may have a carboxymethyl cellulose content of greater than about 4 percent by weight on a dry weight basis.

[0162] The aerosol-generating film may have a carboxymethyl cellulose content of less than about

15 percent by weight on a dry weight basis. The aerosol-generating film may have a carboxymethyl cellulose content of less than about 12 percent by weight on a dry weight basis. The aerosol-generating film may have a carboxymethyl cellulose content of less than about 10 percent by weight on a dry weight basis.

[0163] For example, the aerosol-generating film may have a carboxymethyl cellulose content of between 1 percent and 15 percent by weight, or between 2 percent and 12 percent by weight, or between 4 percent and 10 percent by weight on a dry weight basis.

[0164] The aerosol-generating film preferably comprises nicotine.

[0165] As used herein with reference to the invention, the term “nicotine” is used to describe nicotine, a nicotine base or a nicotine salt. In embodiments in which the aerosol-generating film comprises a nicotine base or a nicotine salt, the amounts of nicotine recited herein are the amount of free base nicotine or amount of protonated nicotine, respectively.

[0166] The aerosol-generating film may comprise natural nicotine or synthetic nicotine.

[0167] The aerosol-generating film may comprise one or more monoprotic nicotine salts.

[0168] As used herein with reference to the invention, the term “monoprotic nicotine salt” is used to describe a nicotine salt of a monoprotic acid.

[0169] Preferably, the aerosol-generating film comprises at least 0.5 percent by weight of nicotine on a dry weight basis. More preferably, the aerosol-generating film comprises at least 1 percent by weight of nicotine on a dry weight basis. Even more preferably, the aerosol-generating film comprises at least 2 percent by weight of nicotine on a dry weight basis. In addition, or as an alternative, the aerosol-generating film preferably comprises less than 10 percent by weight of nicotine on a dry weight basis. More preferably, the aerosol-generating film comprises less than 8 percent by weight of nicotine on a dry weight basis. More preferably, the aerosol-generating film comprises less than 6 percent by weight of nicotine on a dry weight basis.

[0170] For example, the aerosol-generating film may comprise between 0.5 percent and 10 percent by weight of nicotine, or between 1 percent and 8 percent by weight of nicotine, or between 2 percent and 6 percent by weight of nicotine, on a dry weight basis.

[0171] The aerosol-generating film may be a substantially tobacco-free aerosol-generating film.

[0172] In preferred embodiments, the aerosol-generating film comprises an acid. More preferably, the aerosol-generating film comprises one or more organic acids. Even more preferably, the aerosol-generating film comprises one or more carboxylic acids. In particularly preferred embodiments, the acid is lactic acid, benzoic acid, fumaric acid or levulinic acid.

[0173] Preferably, the aerosol-generating film comprises at least about 0.25 percent by weight of an acid on a dry weight basis. More preferably, the aerosol-generating film comprises at least about 0.5 percent by weight of an acid on a dry weight basis. Even more preferably, the aerosol-generating film comprises at least about 1 percent by weight of an acid on a dry weight basis. In addition, or as an alternative, the aerosol-generating film preferably comprises less than about 3.5 percent by weight of an acid on a dry weight basis. More preferably, the aerosol-generating film comprises less than about 3 percent by weight of an acid on a dry weight basis. Even more preferably, the aerosol-generating film comprises less than about 2.5 percent by weight of an acid on a dry weight basis.

[0174] For example, the aerosol-generating film may comprise between 0.25 percent and 3.5 percent by weight of an acid, or between 0.5 percent and 3 percent by weight of an acid, or between 1 percent and 2.5 percent by weight of an acid, on a dry weight basis.

[0175] The aerosol-generating film may have a thickness from about 0.1 millimetres to about 1 millimetre, more preferably from about 0.1 millimetres to about 0.75 millimetres, even more preferably from about 0.1 millimetres to about 0.5 millimetres. In particularly preferred embodiments, a layer of the film-forming composition is formed that has a thickness from about 50 micrometres to 400 micrometres, more preferably from about 100 micrometres to 200 micrometres.

[0176] The aerosol-generating film may optionally be provided within the second aerosol-

generating segment on a suitable carrier element.

[0177] In alternative embodiments of the invention, the second aerosol-generating substrate may comprise a gel composition that includes nicotine, at least one gelling agent and the aerosol former. The gel composition is preferably substantially tobacco free.

[0178] The preferred weight ranges for nicotine in the gel composition are the same as those defined above in relation to aerosol-generating films.

[0179] The gel composition preferably comprises at least 50 percent by weight of aerosol former, more preferably at least 60 percent by weight, more preferably at least 70 percent by weight of aerosol former, on a dry weight basis. The gel composition may comprise up to 80 percent by weight of aerosol former. The aerosol former in the gel composition is preferably glycerol.

[0180] The gel composition preferably includes at least one gelling agent. Preferably, the gel composition includes a total amount of gelling agents in a range from about 0.4 percent by weight to about 10 percent by weight. More preferably, the composition includes the gelling agents in a range from about 0.5 percent by weight to about 8 percent by weight. More preferably, the composition includes the gelling agents in a range from about 1 percent by weight to about 6 percent by weight. More preferably, the composition includes the gelling agents in a range from about 2 percent by weight to about 4 percent by weight. More preferably, the composition includes the gelling agents in a range from about 2 percent by weight to about 3 percent by weight.

[0181] The term “gelling agent” refers to a compound that homogeneously, when added to a 50 percent by weight water/50 percent by weight glycerol mixture, in an amount of about 0.3 percent by weight, forms a solid medium or support matrix leading to a gel. Gelling agents include, but are not limited to, hydrogen-bond crosslinking gelling agents, and ionic crosslinking gelling agents.

[0182] The term “hydrogen-bond crosslinking gelling agent” refers to a gelling agent that forms non-covalent crosslinking bonds or physical crosslinking bonds via hydrogen bonding. Hydrogen bonding is a type of electrostatic dipole-dipole attraction between molecules, not a covalent bond to a hydrogen atom. It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F atom and another very electronegative atom.

[0183] The hydrogen-bond crosslinking gelling agent may include one or more of a galactomannan, gelatin, agarose, or konjac gum, or agar. The hydrogen-bond crosslinking gelling agent may preferably include agar.

[0184] The term “ionic crosslinking gelling agent” refers to a gelling agent that forms non-covalent crosslinking bonds or physical crosslinking bonds via ionic bonding. Ionic crosslinking involves the association of polymer chains by noncovalent interactions. A crosslinked network is formed when multivalent molecules of opposite charges electrostatically attract each other giving rise to a crosslinked polymeric network.

[0185] The ionic crosslinking gelling agent may include low acyl gellan, pectin, kappa carrageenan, iota carrageenan or alginate. The ionic crosslinking gelling agent may preferably include low acyl gellan.

[0186] The gelling agent may include one or more biopolymers. The biopolymers may be formed of polysaccharides.

[0187] Biopolymers include, for example, gellan gums (native, low acyl gellan gum, high acyl gellan gums with low acyl gellan gum being preferred), xanthan gum, alginates (alginic acid), agar, guar gum, and the like. The composition may preferably include xanthan gum. The composition may include two biopolymers. The composition may include three biopolymers. The composition may include the two biopolymers in substantially equal weights. The composition may include the three biopolymers in substantially equal weights.

[0188] The gel composition may further include a viscosifying agent. The viscosifying agent combined with the hydrogen-bond crosslinking gelling agent and the ionic crosslinking gelling agent appears to surprisingly support the solid medium and maintain the gel composition even when the gel composition comprises a high level of glycerol.

[0189] The term “viscosifying agent” refers to a compound that, when added homogeneously into a 25° C., 50 percent by weight water/50 percent by weight glycerol mixture, in an amount of 0.3 percent by weight, increases the viscosity without leading to the formation of a gel, the mixture staying or remaining fluid. Preferably the viscosifying agent refers to a compound that when added homogeneously into a 25° C. 50 percent by weight water/50 percent by weight glycerol mixture, in an amount of 0.3 percent by weight, increases the viscosity to at least 50 cPs, preferably at least 200 cPs, preferably at least 500 cPs, preferably at least 1000 cPs at a shear rate of 0.1 s.sup.-1, without leading to the formation of a gel, the mixture staying or remaining fluid. Preferably the viscosifying agent refers to a compound that when added homogeneously into a 25° C. 50 percent by weight water/50 percent by weight glycerol mixture, in an amount of 0.3 percent by weight, increases the viscosity at least 2 times, or at least 5 times, or at least 10 times, or at least 100 times higher than before addition, at a shear rate of 0.1 s.sup.-1, without leading to the formation of a gel, the mixture staying or remaining fluid.

[0190] The viscosity values recited herein can be measured using a Brookfield RVT viscometer rotating a disc type RV #2 spindle at 25° C. at a speed of 6 revolutions per minute (rpm).

[0191] The gel composition preferably includes the viscosifying agent in a range from about 0.2 percent by weight to about 5 percent by weight. Preferably the composition includes the viscosifying agent in a range from about 0.5 percent by weight to about 3 percent by weight. Preferably the composition includes the viscosifying agent in a range from about 0.5 percent by weight to about 2 percent by weight. Preferably the composition includes the viscosifying agent in a range from about 1 percent by weight to about 2 percent by weight.

[0192] The viscosifying agent may include one or more of xanthan gum, carboxymethyl-cellulose, microcrystalline cellulose, methyl cellulose, gum Arabic, guar gum, lambda carrageenan, or starch. The viscosifying agent may preferably include xanthan gum.

[0193] The gel composition may further include a divalent cation. Preferably the divalent cation includes calcium ions, such as calcium lactate in solution. Divalent cations (such as calcium ions) may assist in the gel formation of compositions that include gelling agents such as the ionic crosslinking gelling agent, for example. The ion effect may assist in the gel formation. The divalent cation may be present in the gel composition in a range from about 0.1 to about 1 percent by weight, or about 0.5 percent by weight t.

[0194] The gel composition may further include an acid. The acid may comprise a carboxylic acid. The carboxylic acid may include a ketone group. Preferably the carboxylic acid may include a ketone group having less than about 10 carbon atoms, or less than about 6 carbon atoms or less than about 4 carbon atoms, such as levulinic acid or lactic acid. Preferably this carboxylic acid has three carbon atoms (such as lactic acid).

[0195] The gel composition preferably comprises some water. The gel composition is more stable when the composition comprises some water. Preferably the gel composition comprises at least about 1 percent by weight, or at least about 2 percent by weight, or at least about 5 percent by weight of water. Preferably the gel composition comprises at least about 10 percent by weight or at least about 15 percent by weight water.

[0196] Preferably the gel composition comprises between about 8 percent by weight to about 32 percent by weight water. Preferably the gel composition comprises from about 15 percent by weight to about 25 percent by weight water. Preferably the gel composition comprises from about 18 percent by weight to about 22 percent by weight water. Preferably the gel composition comprises about 20 percent by weight water.

[0197] Preferably, where a gel composition is used, the second aerosol-generating substrate comprises a porous medium loaded with the gel composition. Advantages of a porous medium loaded with the gel composition is that the gel composition is retained within the porous medium, and this may aid manufacturing, storage or transport of the gel composition. It may assist in keeping the desired shape of the gel composition, especially during manufacture, transport, or use.

[0198] The term “porous” is used herein to refer to a material that provides a plurality of pores or openings that allow the passage of air through the material.

[0199] The porous medium may be any suitable porous material able to hold or retain the gel composition. Ideally the porous medium can allow the gel composition to move within it. In specific embodiments the porous medium comprises natural materials, synthetic, or semi-synthetic, or a combination thereof. In specific embodiments the porous medium comprises sheet material, foam, or fibres, for example loose fibres; or a combination thereof. In specific embodiments the porous medium comprises a woven, non-woven, or extruded material, or combinations thereof. Preferably the porous medium comprises, cotton, paper, viscose, PLA, or cellulose acetate, or combinations thereof. Preferably the porous medium comprises a sheet material, for example, cotton or cellulose acetate. In a particularly preferred embodiment, the porous medium comprises a sheet made from cotton fibres.

[0200] The porous medium used in the present invention may be crimped or shredded. In preferred embodiments, the porous medium is crimped. In alternative embodiments the porous medium comprises shredded porous medium. The crimping or shredding process can be before or after loading with the gel composition.

[0201] In specific embodiments the sheet material is a composite material. Preferably the sheet material is porous. The sheet material may aid manufacture of the tubular element comprising a gel. The sheet material may aid introducing an active agent to the tubular element comprising a gel. The sheet material may help stabilise the structure of the tubular element comprising a gel. The sheet material may assist transport or storage of the gel. Using a sheet material enables, or aids, adding structure to the porous medium for example by crimping of the sheet material.

[0202] The porous medium may be a thread. The thread may comprise for example cotton, paper or acetate tow. The thread may also be loaded with gel like any other porous medium. An advantage of using a thread as the porous medium is that it may aid ease of manufacturing.

[0203] The thread may be loaded with gel by any known means. The thread may be simply coated with gel, or the thread may be impregnated with gel. In the manufacture, the threads may be impregnated with gel and stored ready for use to be included in the assembly of a tubular element.

[0204] The porous medium loaded with the gel composition is preferably provided within a tubular element that forms a part of the aerosol-generating article. Ideally the tubular element may be longer in longitudinal length than in width but not necessarily as it may be one part of a multi-component item that ideally will be longer in its longitudinal length than its width. Typically, the tubular element is cylindrical but not necessarily. For example, the tubular element may have an oval, polygonal like triangular or rectangular or random cross section.

[0205] The second aerosol-generating substrate may comprise an aerosol-generating film. The aerosol-generating film may comprise a cellulosic based film forming agent, nicotine and glycerol. The aerosol-generating film may have a glycerol content of at least 40 percent by weight.

[0206] The first aerosol-generating substrate and the second aerosol-generating substrate comprise at least one aerosol former, the aerosol former content of the second aerosol-generating substrate may be greater than the aerosol former content of the first aerosol-generating substrate.

[0207] The provision of a lower aerosol former content in the first aerosol-generating substrate may advantageously provide improved aerosol delivery of the aerosol-generating article. The first aerosol-generating substrate, which has a lower level of aerosol former, will heat up faster than it would if it had a higher aerosol former content due to the lower thermal inertia. This is advantageous since the majority of the first aerosol-generating substrate is not located in the heating zone and so will be heated to a lower temperature than the second aerosol-generating substrate. Providing a lower aerosol former content in the first aerosol-generating substrate may advantageously allow for effective aerosol-generation in the first aerosol-generating substrate despite the indirect heating most of the aerosol-generating substrate will receive. By contrast, the second aerosol-generating substrate which is mostly located within the heating zone, can include a

higher aerosol former content as the second aerosol-generating substrate is more efficiently heated. The higher aerosol former content in the second aerosol-generating substrate may advantageously allow for sustained aerosol delivery over the full duration of the use of the aerosol-generating article.

[0208] Upon volatilisation, an aerosol former can convey other vaporised compounds released from the aerosol-generating substrate upon heating, such as nicotine and flavourants, in an aerosol.

[0209] Suitable aerosol formers for inclusion in the first and second aerosol-generating substrates are known in the art and include, but are not limited to: polyhydric alcohols, such as triethylene glycol, propylene glycol, 1,3-butanediol and glycerol; esters of polyhydric alcohols, such as glycerol mono-, di- or triacetate; and aliphatic esters of mono-, di- or polycarboxylic acids, such as dimethyl dodecanedioate and dimethyl tetradecanedioate.

[0210] The first and second aerosol-generating substrates may comprise the same aerosol former (or aerosol formers) as each other, or different aerosol formers may be used.

[0211] The first aerosol-generating substrate preferably has an aerosol former content of no more than 30 percent by weight on a dry weight basis. More preferably, the first aerosol-generating substrate has an aerosol former content of no more than 25 percent by weight on a dry weight basis. More preferably, the first aerosol-generating substrate has an aerosol former content of no more than 20 percent by weight on a dry weight basis.

[0212] Preferably, the first aerosol-generating substrate has an aerosol former content of at least 5 percent by weight on a dry weight basis. More preferably, the first aerosol-generating substrate has an aerosol former content of at least 10 percent by weight on a dry weight basis. More preferably, the first aerosol-generating substrate has an aerosol former content of at least 12 percent by weight on a dry weight basis. More preferably, the first aerosol-generating substrate has an aerosol former content of at least 15 percent by weight on a dry weight basis.

[0213] For example, the aerosol former content of the first aerosol-generating substrate may be between 5 percent and 30 percent by weight, or between 10 percent and 25 percent by weight, or between 12 percent and 20 percent by weight, or between about 15 percent and about 20 percent by weight, on a dry weight basis.

[0214] Preferably, the first aerosol-generating substrate comprises glycerol as aerosol former. For example, the first aerosol-generating substrate may comprise between 5 percent and 30 percent by weight of glycerol, or between 10 percent and 25 percent by weight of glycerol, or between 12 percent and 20 percent by weight of glycerol, or between 15 percent and 20 percent by weight of glycerol, on a dry weight basis.

[0215] The second aerosol-generating substrate preferably has a higher aerosol former content than the first aerosol-generating substrate.

[0216] Preferably, the second aerosol-generating substrate has an aerosol former content of at least 40 percent by weight on a dry weight basis. More preferably, the second aerosol-generating substrate has an aerosol former content of at least 45 percent by weight on a dry weight basis. More preferably, the second aerosol-generating substrate has an aerosol former content of at least 50 percent by weight on a dry weight basis.

[0217] Preferably, the second aerosol-generating substrate has an aerosol former content of no more than 80 percent by weight on a dry weight basis. More preferably, the second aerosol-generating substrate has an aerosol former content of no more than 75 percent by weight on a dry weight basis. More preferably, the second aerosol-generating substrate has an aerosol former content of no more than 70 percent by weight on a dry weight basis.

[0218] For example, the aerosol former content of the second aerosol-generating substrate may be between 40 percent and 80 percent by weight, or between 45 percent and 75 percent by weight, or between 50 percent and 70 percent by weight, on a dry weight basis.

[0219] Preferably, the second aerosol-generating substrate comprises glycerol as aerosol former. For example, the second aerosol-generating substrate may comprise between 40 percent and 80

percent by weight of glycerol, or between 45 percent and 75 percent by weight of glycerol, or between 50 percent and 70 percent by weight of glycerol, or between 15 percent and 20 percent by weight of glycerol, on a dry weight basis.

[0220] Preferably, the aerosol former content of the second aerosol-generating substrate is at least 15 percent higher than the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. For example, if the first aerosol-generating substrate had an aerosol former content of 10 percent on a dry weight basis, the second aerosol-generating substrate may have an aerosol former content of no less than 25 percent on a dry weight basis.

[0221] More preferably, the aerosol former content of the second aerosol-generating substrate is at least 20 percent higher than the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. More preferably, the aerosol former content of the second aerosol-generating substrate is at least 25 percent higher than the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. The aerosol former content of the second aerosol-generating substrate may be up to 60 percent higher than the aerosol former content of the first aerosol-generating substrate.

[0222] Preferably, the aerosol former content of the second aerosol-generating substrate is at least 1.2 times the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. More preferably, the aerosol former content of the second aerosol-generating substrate is at least 1.5 times the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. More preferably, the aerosol former content of the second aerosol-generating substrate is at least twice the aerosol former content of the first aerosol-generating substrate, on a dry weight basis. The aerosol former content of the second aerosol-generating substrate may be up to 4 times the aerosol former content of the first aerosol-generating substrate.

[0223] In preferred embodiments of the invention, the density of the second aerosol-generating substrate is at least 1.2 times the density of the first aerosol-generating substrate, the aerosol former content of the second aerosol-generating substrate is at least twice the aerosol former content of the first aerosol-generating substrate on a dry weight basis and the ratio of the length of the first aerosol-generating segment to the second aerosol-generating segment is no more than 0.5.

[0224] The aerosol-generating article may further comprise a third aerosol-generating segment, downstream of the first aerosol-generating segment and downstream of the second aerosol-generating segment, and comprising a third aerosol-generating substrate. For example, the aerosol-generating article may comprise a third aerosol-generating segment abutting the downstream end of the second aerosol-generating segment. The inclusion of a third aerosol-generating segment downstream of the second aerosol-generating segment may further enhance the aerosol delivery from the aerosol-generating articles according to the invention. When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the entire length of the third aerosol-generating segment may be disposed within the heating zone. When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the third aerosol-generating segment may be disposed only partially within the heating zone. When the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device, the entire length of the third aerosol-generating segment may be disposed outside the heating zone.

[0225] The advantage of providing a third aerosol-generating segment comprising a third aerosol-generating substrate downstream of the first and second aerosol-generating segments is that it may allow the aerosol-generating segment to be tailored to deliver an aerosol during the earlier puffs of the use of the aerosol-generating system. For example, the third aerosol-generating substrate may have a relatively low aerosol former content or a relatively low density, or both. For example, the third aerosol-generating substrate preferably has an aerosol former content and density within the ranges defined above for the first aerosol-generating substrate. The teaching provided above in relation to the first aerosol-generating substrate also applies in relation to the third aerosol-generating

substrate.

[0226] This may advantageously allow the third aerosol-generating substrate to heat up faster due to the lower thermal inertia and begin to produce a measurable amount of aerosol within a relatively short time. Since the third aerosol-generating segment is located downstream of both the first and second aerosol-generating segments, the aerosol generated by the third aerosol-generating substrate is delivered efficiently to a user with minimal self filtration. This further improves aerosol delivery during the earlier puffs of the use of the aerosol-generating system.

[0227] The third aerosol-generating segment may have any length. The third aerosol-generating segment may have a length of at least 2 millimetres. For example, the third aerosol-generating segment may have a length of at least 3 millimetres or at least 4 millimetres.

[0228] The third aerosol-generating segment may have a length of no more than 8 millimetres. For example, the third aerosol-generating segment may have a length of no more than 7 millimetres or no more than 6 millimetres.

[0229] The third aerosol-generating segment may have a length of between 2 millimetres and 8 millimetres, between 3 millimetres and 7 millimetres, or between 4 millimetres and 6 millimetres.

[0230] The third aerosol-generating segment may have a length of between 1 millimetre and 6 millimetres, between 2 millimetres and 5 millimetres, or between 3 millimetres and 4 millimetres.

[0231] The third aerosol-generating segment may have a length of about 5 millimetres. The third aerosol-generating segment may have a length of about 3.5 millimetres.

[0232] In some embodiments, the first, second, and third aerosol-generating segments are arranged along the longitudinal axis of the aerosol-generating article. The first aerosol-generating segment may be disposed furthest upstream, the third aerosol-generating segment may be disposed furthest downstream. The second aerosol-generating segment may be disposed between the first and third aerosol-generating segments. The downstream end of the first aerosol-generating segment may abut the upstream end of the second aerosol-generating segment. The downstream end of the second aerosol-generating segment may abut the upstream end of the third aerosol-generating segment.

[0233] The combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be at least 10 millimetres. For example, the combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be at least 12 millimetres, at least 14 millimetres, or at least 16 millimetres.

[0234] The combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be no more than 24 millimetres. For example, the combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be no more than 22 millimetres, no more than 20 millimetres, or no more than 18 millimetres.

[0235] The combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be between 10 millimetres and 24 millimetres. For example, the combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be between 12 millimetres and 22 millimetres, between 14 millimetres and 20 millimetres, or between 16 millimetres and 18 millimetres.

[0236] The combined length of the first aerosol-generating segment, the second aerosol-generating segment, and the third aerosol-generating segment may be about 17 millimetres.

[0237] The aerosol-generating segments may have an external diameter that is approximately equal to the external diameter of the aerosol-generating article.

[0238] Preferably, the external diameter of the aerosol-generating segments is between 5 millimetres and 12 millimetres, more preferably between 6 millimetres and 10 millimetres, more preferably between 7 millimetres and 8 millimetres. In some embodiments, the external diameter of the aerosol-generating segments may be less than 7 millimetres, for example, between 5

millimetres and 7 millimetres, or between 6 millimetres and 7 millimetres.

[0239] The first aerosol-generating segment preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. Preferably, the external diameter of the first aerosol-generating segment is substantially constant along the length of the first aerosol-generating segment.

[0240] Preferably, the external diameter of the first aerosol-generating segment is between 5 millimetres and 12 millimetres, more preferably between 6 millimetres and 10 millimetres, more preferably between 7 millimetres and 8 millimetres. In some embodiments, the external diameter of the first aerosol-generating segment may be less than 7 millimetres, for example, between 5 millimetres and 7 millimetres, or between 6 millimetres and 7 millimetres.

[0241] The second aerosol-generating segment preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. Preferably, the external diameter of the second aerosol-generating segment is substantially constant along the length of the first aerosol-generating segment.

[0242] Preferably, the external diameter of the second aerosol-generating segment is between 5 millimetres and 12 millimetres, more preferably between 6 millimetres and 10 millimetres, more preferably between 7 millimetres and 8 millimetres. In some embodiments, the external diameter of the second aerosol-generating segment may be less than 7 millimetres, for example, between 5 millimetres and 7 millimetres, or between 6 millimetres and 7 millimetres.

[0243] The third aerosol-generating segment preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. Preferably, the external diameter of the third aerosol-generating segment is substantially constant along the length of the third aerosol-generating segment.

[0244] Preferably, the external diameter of the third aerosol-generating segment is between 5 millimetres and 12 millimetres, more preferably between 6 millimetres and 10 millimetres, more preferably between 7 millimetres and 8 millimetres. In some embodiments, the external diameter of the third aerosol-generating segment may be less than 7 millimetres, for example, between 5 millimetres and 7 millimetres, or between 6 millimetres and 7 millimetres.

[0245] Preferably, the first aerosol-generating segment, the second aerosol-generating segment and the third aerosol-generating segment (where present) have substantially the same external diameter as each other.

[0246] The average cross-sectional area of the first aerosol-generating segment is preferably at least 50 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 80 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 90 percent of the average cross-sectional area of the aerosol-generating article.

[0247] The cross-sectional area of the first aerosol-generating segment at the upstream end thereof is preferably at least 50 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 80 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 90 percent of the average cross-sectional area of the aerosol-generating article.

[0248] The average cross-sectional area of the second aerosol-generating segment is preferably at least 50 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 80 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 90 percent of the average cross-sectional area of the aerosol-generating article.

[0249] The cross-sectional area of the second aerosol-generating segment at the downstream end thereof is preferably at least 50 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 80 percent of the average cross-sectional area of the aerosol-generating article, more preferably at least 90 percent of the average cross-sectional area of the

aerosol-generating article.

[0250] The cross-sectional area of the first aerosol-generating segment at the upstream end thereof is substantially the same as the cross-section area of the second aerosol-generating segment at the downstream end thereof.

[0251] The aerosol-generating articles of the present invention preferably comprise an upstream element located upstream of and adjacent to the first aerosol-generating segment. The upstream element advantageously prevents direct physical contact with the upstream end of the aerosol-generating segment. For example, where the aerosol-generating substrate comprises a susceptor element, the upstream element may prevent direct physical contact with the upstream end of the susceptor element. This helps to prevent the displacement or deformation of the susceptor element during handling or transport of the aerosol-generating article. This in turn helps to secure the form and position of the susceptor element. Furthermore, the presence of an upstream element helps to prevent any loss of the substrate, which may be advantageous, for example, if the substrate contains particulate plant material.

[0252] Where the first aerosol-generating segment comprises shredded tobacco, such as tobacco cut filler, the upstream section or element thereof may additionally help to prevent the loss of loose particles of tobacco from the upstream end of the article. This may be particularly important when the shredded tobacco has a relatively low density, for example.

[0253] The upstream section, or upstream element thereof, may also additionally provide a degree of protection to the aerosol-generating substrate during storage, as it covers at least to some extent the upstream end of the aerosol-generating substrate, which may otherwise be exposed.

[0254] For aerosol-generating articles that are intended to be inserted into a heating chamber in an aerosol-generating device such that the aerosol-generating substrate can be externally heated within the heating chamber, the upstream section, or upstream element thereof, may advantageously facilitate the insertion of the upstream end of the article into the heating chamber. The inclusion of the upstream element may additionally protect the end of the first aerosol-generating segment during the insertion of the article into the heating chamber such that the risk of damage to the substrate is minimised.

[0255] An upstream element may be a porous plug element. Preferably, an upstream element has a porosity of at least 50 percent in the longitudinal direction of the aerosol-generating article. More preferably, an upstream element has a porosity of between 50 percent and 90 percent in the longitudinal direction. The porosity of an upstream element in the longitudinal direction is defined by the ratio of the cross-sectional area of material forming the upstream element and the internal cross-sectional area of the aerosol-generating article at the position of the upstream element.

[0256] An upstream element may be made of a porous material or may comprise a plurality of openings. This may, for example, be achieved through laser perforation. Preferably, the plurality of openings is distributed homogeneously over the cross-section of the upstream element.

[0257] The porosity or permeability of an upstream element may advantageously be designed in order to provide an aerosol-generating article with a particular overall resistance to draw (RTD) without substantially impacting the filtration provided by other portions of the article.

[0258] An upstream element may be formed from a material that is impermeable to air. In such embodiments, the aerosol-generating article may be configured such that air flows into at least one of the first aerosol-generating segment and the second aerosol-generating segment through suitable ventilation means provided in a wrapper.

[0259] In certain preferred embodiments of the invention, it may be desirable to minimise the RTD of an upstream element. For example, this may be the case for articles that are intended to be inserted the heating chamber of an aerosol-generating device such that the aerosol-generating substrate is externally heated, as described herein. For such articles, it is desirable to provide the article with as low an RTD as possible, so that the majority of the RTD experience by the consumer is provided by the aerosol-generating device and not the article.

[0260] The overall RTD of the aerosol-generating article may be at least 10 millimetres H₂O. For example, the overall RTD of the aerosol-generating article may be at least 20 millimetres H₂O, at least 30 millimetres H₂O, at least 35 millimetres H₂O, or at least 40 millimetres H₂O.

[0261] The overall RTD of the aerosol-generating article may be no more than 70 millimetres H₂O. For example, the overall RTD of the aerosol-generating article may be no more than 60 millimetres H₂O, no more than 55 millimetres H₂O, no more than 50 millimetres H₂O, or no more than 45 millimetres H₂O.

[0262] The overall RTD of the aerosol-generating article may be between 10 millimetres H₂O and 70 millimetres H₂O. For example, the overall RTD of the aerosol-generating article may be between 20 millimetres H₂O and 60 millimetres H₂O, between 30 millimetres H₂O and 55 millimetres H₂O, between 35 millimetres H₂O and 50 millimetres H₂O, or between 40 millimetres H₂O and 45 millimetres H₂O.

[0263] The overall RTD of the aerosol-generating article may be between 40 millimetres H₂O and 60 millimetres H₂O, between 35 millimetres H₂O and 40 millimetres H₂O, between 45 millimetres H₂O and 50 millimetres H₂O, or between 55 millimetres H₂O and 65 millimetres H₂O.

[0264] The overall RTD of the aerosol-generating article may be about 38 millimetres H₂O, about 48 millimetres H₂O, or about 60 millimetres H₂O.

[0265] Aerosol-generating articles according to the present invention preferably further comprise a downstream section located downstream of the first aerosol-generating segment, the second aerosol-generating segment, and where present the third aerosol-generating segment. The downstream section is preferably located immediately downstream of the second aerosol-generating segment, or where present the third aerosol-generating segment. The downstream section of the aerosol-generating article preferably extends between the second aerosol-generating segment, or where present the third aerosol-generating segment and the downstream end of the aerosol-generating article. The downstream section may comprise one or more elements, each of which will be described in more detail within the present disclosure.

[0266] A length of the downstream section may be at least 20 millimetres. A length of the downstream section may be at least 25 millimetres. A length of the downstream section may be at least 30 millimetres.

[0267] A length of the downstream section may be less than 70 millimetres. A length of the downstream section may be equal to or less than 60 millimetres. A length of the downstream section may be equal to or less than 50 millimetres.

[0268] For example, a length of the downstream section may be between 20 millimetres and 70 millimetres, or between 25 millimetres and 60 millimetres, or between 30 millimetres and 50 millimetres.

[0269] Providing a relatively long downstream section ensures that a suitable length of the aerosol-generating article protrudes from an aerosol-generating device when the article is received therein. Such a suitable protrusion length facilitates the ease of insertion and extraction of the article from the device, which also ensures that the upstream portions of the article are suitably inserted into the device with reduced risk of damage, particularly during insertion.

[0270] A ratio between a length of the downstream section and an overall length of the aerosol-generating article may be less than 0.80. More preferably, a ratio between a length of the downstream section and an overall length of the aerosol-generating article may be less than 0.75. Even more preferably, a ratio between a length of the downstream section and an overall length of the aerosol-generating article may be less than 0.70.

[0271] A ratio between a length of the downstream section and an overall length of the aerosol-generating article may be at least 0.30. Preferably, a ratio between a length of the downstream section and an overall length of the aerosol-generating article may be at least 0.40. More preferably, a ratio between a length of the downstream section and an overall length of the aerosol-generating article may be at least 0.50.

[0272] In some embodiments, a ratio between a length of the downstream section and an overall length of the aerosol-generating article is from 0.30 to 0.80, preferably from 0.40 to 0.75, more preferably from 0.50 to 0.70.

[0273] The downstream section of an aerosol-generating article according to the present invention preferably comprises a hollow tubular cooling element provided downstream of the second aerosol-generating segment, or where present the third aerosol-generating segment. The hollow tubular cooling element may advantageously provide an aerosol-cooling element for the aerosol-generating article.

[0274] The hollow tubular cooling element may be provided immediately downstream of the second aerosol-generating segment, or where present the third aerosol-generating segment. In other words, the hollow tubular cooling element may abut a downstream end of the second aerosol-generating segment, or where present the third aerosol-generating segment. The hollow tubular cooling element may define an upstream end of the downstream section of the aerosol-generating article. The downstream end of the aerosol-generating article may coincide with the downstream end of the downstream section. In some embodiments, the downstream section of the aerosol-generating article comprises a single hollow tubular element. In other words, the downstream section of the aerosol-generating article may comprise only one hollow tubular element. In other embodiments, the downstream section comprises two or more hollow tubular elements, as described below.

[0275] As used throughout the present disclosure, the term “hollow tubular element” denotes a generally elongate element defining a lumen or airflow passage along a longitudinal axis thereof. In particular, the term “tubular” will be used in the following with reference to a tubular element having a substantially cylindrical cross-section and defining at least one airflow conduit establishing an uninterrupted fluid communication between an upstream end of the tubular element and a downstream end of the tubular element. However, it will be understood that alternative geometries (for example, alternative cross-sectional shapes) of the tubular element may be possible.

[0276] In the context of the present invention, a hollow tubular cooling element provides an unrestricted flow channel. This means that the hollow tubular cooling element provides a negligible level of resistance to draw (RTD). The term “negligible level of RTD” is used to describe an RTD of less than 1 millimetres H.sub.2O per 10 millimetres of length of the hollow tubular cooling element, preferably less than 0.4 millimetres H.sub.2O per 10 millimetres of length of the hollow tubular cooling element, more preferably less than 0.1 millimetres H.sub.2O per 10 millimetres of length of the hollow tubular cooling element.

[0277] The RTD of a hollow tubular cooling element is preferably less than or equal to 10 millimetres H.sub.2O. More preferably, the RTD of a hollow tubular cooling element is less than or equal to 5 millimetres H.sub.2O. Even more preferably, the RTD of a hollow tubular cooling element is less than or equal to 2.5 millimetres H.sub.2O. Even more preferably, the RTD of the hollow tubular cooling element is less than or equal to 2 millimetres H.sub.2O. Even more preferably, the RTD of the hollow tubular cooling element is less than or equal to 1 millimetre H.sub.2O.

[0278] The RTD of a hollow tubular cooling element may be at least 0 millimetres H.sub.2O, or at least 0.25 millimetres H.sub.2O or at least 0.5 millimetres H.sub.2O or at least 1 millimetre H.sub.2O.

[0279] In aerosol-generating articles in accordance with the present invention the overall RTD of the article depends essentially on the RTD of the aerosol-generating segments and optionally on the RTD of the downstream and/or upstream elements. This is because the hollow tubular cooling element is substantially empty and, as such, substantially only marginally contribute to the overall RTD of the aerosol-generating article.

[0280] The flow channel should therefore be free from any components that would obstruct the flow of air in a longitudinal direction. Preferably, the flow channel is substantially empty and

particularly preferably the flow channel is empty.

[0281] As will be described in greater detail within the present disclosure, the aerosol-generating article may comprise a ventilation zone at a location along the downstream section. In some embodiments, the aerosol-generating article may comprise a ventilation zone at a location along the hollow tubular cooling element. Such, or any, ventilation zone may extend through the peripheral wall of the hollow tubular cooling element. As such, fluid communication is established between the flow channel internally defined by the hollow tubular cooling element and the outer environment. The ventilation zone is further described within the present disclosure.

[0282] Preferably, the length of the hollow tubular cooling element is at least 15 millimetres. More preferably, the length of the hollow tubular cooling element is at least 20 millimetres. The length of the hollow tubular cooling element may be at least 25 millimetres. More preferably, the length of the hollow tubular cooling element is at least 30 millimetres.

[0283] The length of the hollow tubular cooling element is preferably less than 50 millimetres. More preferably, the length of the hollow tubular cooling element is less than 45 millimetres. More preferably, the length of the hollow tubular cooling element is less than 40 millimetres.

[0284] A relatively long hollow tubular cooling element provides and defines a relatively long internal cavity within the aerosol-generating article and downstream of the second aerosol-generating segment, or where present the third aerosol-generating segment. As discussed in the present disclosure, providing an empty cavity downstream (preferably, immediately downstream) of the aerosol-generating substrate enhances the nucleation of aerosol particles generated by the substrate. Providing a relatively long cavity maximises such nucleation benefits, thereby improving aerosol formation and cooling.

[0285] The thickness of a peripheral wall (in other words, the wall thickness) of the hollow tubular cooling element may be at least 100 micrometres. The wall thickness of the hollow tubular cooling element may be at least 150 micrometres. The wall thickness of the hollow tubular cooling element may be at least 200 micrometres, preferably at least 250 micrometres and even more preferably at least 500 micrometres (or 0.5 millimetres).

[0286] The wall thickness of the hollow tubular cooling element may be less than or equal to 2 millimetres, preferably less than or equal to 1.5 millimetres and even more preferably less than or equal to 1.25 millimetres. The wall thickness of the hollow tubular cooling element may be less than or equal to 1 millimetre. The wall thickness of the hollow tubular cooling element may be less than or equal to 500 micrometres.

[0287] The wall thickness of the hollow tubular cooling element may be between 100 micrometres and 2 millimetres, preferably between 150 micrometres and 1.5 millimetres, even more preferably between 200 micrometres and 1.25 millimetres.

[0288] The wall thickness of the hollow tubular cooling element may preferably be 250 micrometres (0.25 millimetres).

[0289] At the same time, keeping the thickness of the peripheral wall of the hollow tubular cooling element relatively low ensures that the overall internal volume of the hollow tubular cooling element—which is made available for the aerosol to begin the nucleation process as soon as the aerosol components leave the aerosol-generating segments—and the cross-sectional surface area of the hollow tubular cooling element are effectively maximised, whilst at the same time ensuring that the hollow tubular cooling element has the necessary structural strength to prevent a collapse of the aerosol-generating article as well as to provide some support to the aerosol-generating segments, and that the RTD of the hollow tubular cooling element is minimised. Greater values of cross-sectional surface area of the cavity of the hollow tubular cooling element are understood to be associated with a reduced speed of the aerosol stream travelling along the aerosol-generating article, which is also expected to favour aerosol nucleation. Further, it would appear that by utilising a hollow tubular cooling element having a relatively low thickness, it is possible to substantially prevent diffusion of the ventilation air prior to its contacting and mixing with the

stream of aerosol, which is also understood to further favour nucleation phenomena. In practice, by providing a more controllably localised cooling of the stream of volatilised species, it is possible to enhance the effect of cooling on the formation of new aerosol particles.

[0290] The hollow tubular cooling element preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating segments and to the external diameter of the aerosol-generating article.

[0291] The hollow tubular cooling element may have an internal diameter. Preferably, the hollow tubular cooling element may have a constant internal diameter along a length of the hollow tubular cooling element. However, the internal diameter of the hollow tubular cooling element may vary along the length of the hollow tubular cooling element.

[0292] The hollow tubular cooling element may have an internal diameter of at least 2 millimetres. For example, the hollow tubular cooling element may have an internal diameter of at least 3 millimetres, at least 4 millimetres, or at least 5 millimetres.

[0293] The provision of a hollow tubular cooling element having an internal diameter as set out above may advantageously provide sufficient rigidity and strength to the hollow tubular cooling element.

[0294] The hollow tubular cooling element may have an internal diameter of no more than 10 millimetres. For example, the hollow tubular cooling element may have an internal diameter of no more than 9 millimetres, no more than 8 millimetres, or no more than 7 millimetres.

[0295] The provision of a hollow tubular cooling element having an internal diameter as set out above may advantageously reduce the resistance to draw of the hollow tubular cooling element.

[0296] The hollow tubular cooling element may have an internal diameter of between 2 millimetres and 10 millimetres, between 3 millimetres and 9 millimetres, between 4 millimetres and 8 millimetres, or between 5 millimetres and 7 millimetres.

[0297] The hollow tubular cooling element may comprise a paper-based material. The hollow tubular cooling element may comprise at least one layer of paper. The paper may be very rigid paper. The paper may be crimped paper, such as crimped heat resistant paper or crimped parchment paper.

[0298] Preferably, the hollow tubular cooling element may comprise cardboard. The hollow tubular cooling element may be a cardboard tube. The hollow tubular cooling element may be formed from cardboard.

[0299] The hollow tubular cooling element may comprise a polymeric material. For example, the hollow tubular cooling element may comprise a polymeric film. The polymeric film may comprise a cellulosic film. The hollow tubular cooling element may comprise low density polyethylene (LDPE) or polyhydroxyalkanoate (PHA) fibres. The hollow tube may comprise cellulose acetate tow.

[0300] In some embodiments, the aerosol-generating article according to the present invention may comprise a ventilation zone at a location along the downstream section. In more detail, in those embodiments wherein the downstream section comprises a hollow tubular cooling element, the ventilation zone may be provided at a location along the hollow tubular cooling element.

Alternatively, in those embodiments where the downstream section comprises a downstream hollow tubular element, as described below, the ventilation zone may be provided at a location along the downstream hollow tubular element.

[0301] As such, a ventilated cavity is provided downstream of the aerosol-generating segments. This provides several potential technical benefits.

[0302] First of all, the inventors have found that one such ventilated hollow tubular cooling element provides a particularly efficient cooling of the aerosol. Thus, a satisfactory cooling of the aerosol can be achieved even by means of a relatively short downstream section.

[0303] Secondly, the inventors have surprisingly found that such rapid cooling of the volatile species released upon heating the aerosol-generating substrate promotes enhances nucleation of

aerosol particles.

[0304] The ventilation zone may typically comprise a plurality of perforations through the peripheral wall of the hollow tubular cooling element. Preferably, the ventilation zone comprises at least one circumferential row of perforations. In some embodiments, the ventilation zone may comprise two circumferential rows of perforations. For example, the perforations may be formed online during manufacturing of the aerosol-generating article. Preferably, each circumferential row of perforations comprises from 8 to 30 perforations.

[0305] An aerosol-generating article in accordance with the present invention may have a ventilation level of at least 25 percent.

[0306] The term “ventilation level” is used throughout the present specification to denote a volume ratio between of the airflow admitted into the aerosol-generating article via the ventilation zone (ventilation airflow) and the sum of the aerosol airflow and the ventilation airflow. The greater the ventilation level, the higher the dilution of the aerosol flow delivered to the consumer. The aerosol-generating article preferably has a ventilation level of at least 25 percent, more preferably at least 30 percent, even more preferably at least 40 percent, even more preferably at least 50 percent.

[0307] An aerosol-generating article in accordance with the present invention may have a ventilation level of up to 90 percent. Preferably, an aerosol-generating article in accordance with the present invention has a ventilation level of less than or equal to 80 percent, more preferably less than or equal to 70 percent, even more preferably less than or equal to 60 percent.

[0308] Without wishing to be bound by theory, the inventors have found that the temperature drop caused by the admission of cooler, external air into the hollow tubular cooling element via the ventilation zone may have an advantageous effect on the nucleation and growth of aerosol particles.

[0309] As discussed in the present disclosure, the downstream section may comprise a downstream filter segment. The downstream filter segment may extend to a downstream end of the downstream section. The downstream filter segment may be located at the downstream end of the aerosol-generating article. The downstream end of the downstream filter segment may define the downstream end of the aerosol-generating article.

[0310] The downstream filter segment may be located downstream of a hollow tubular cooling element, which is described above. The downstream filter segment may extend between the hollow tubular cooling element and the downstream end of the aerosol-generating article.

[0311] The downstream filter segment is preferably a solid plug, which may also be described as a ‘plain’ plug and is non-tubular. The filter segment therefore preferably has a substantially uniform transverse cross section.

[0312] The downstream filter segment is preferably formed of a fibrous filtration material. The fibrous filtration material may be for filtering the aerosol that is generated from the aerosol-generating substrate. Suitable fibrous filtration materials would be known to the skilled person. Particularly preferably, the at least one downstream filter segment comprises a cellulose acetate filter segment formed of cellulose acetate tow.

[0313] In certain preferred embodiments, the downstream section includes a single downstream filter segment. In alternative embodiments, the downstream section includes two or more downstream filter segments axially aligned in an abutting end to end relationship with each other.

[0314] The downstream filter segment may optionally comprise a flavourant, which may be provided in any suitable form. For example, the downstream filter segment may comprise one or more capsules, beads or granules of a flavourant, or one or more flavour loaded threads or filaments.

[0315] Preferably, the downstream filter segment has a low particulate filtration efficiency.

[0316] Preferably, the downstream filter segment is circumscribed by a plug wrap. Preferably, the downstream filter segment is unventilated such that air does not enter the aerosol-generating article along the downstream filter segment.

[0317] The downstream filter segment is preferably connected to one or more of the adjacent upstream components of the aerosol-generating article by means of a tipping wrapper.

[0318] The downstream filter segment preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. The diameter of a downstream filter segment may be substantially the same as the external diameter of the hollow tubular cooling element.

[0319] The external diameter of the downstream filter segment may be between 5 millimetres and 12 millimetres. The diameter of the downstream filter segment may be between 6 millimetres and 10 millimetres, between 7 millimetres and 8 millimetres. In certain embodiments, the diameter of the downstream filter segment may be less than 7 millimetres, for example, between 5 millimetres and 7 millimetres, or between 6 millimetres and 7 millimetres.

[0320] Unless otherwise specified, the resistance to draw (RTD) of a component or the aerosol-generating article is measured in accordance with ISO 6565-2015. The RTD refers the pressure required to force air through the full length of a component. The terms “pressure drop” or “draw resistance” of a component or article may also refer to the “resistance to draw”. Such terms generally refer to the measurements in accordance with ISO 6565-2015 are normally carried out at under test at a volumetric flow rate of 17.5 millilitres per second at the output or downstream end of the measured component at a temperature of 22 degrees Celsius, a pressure of 101 kPa (about 760 Torr) and a relative humidity of 60%. Conditions for smoking and smoking machine specifications are set out in ISO Standard 3308 (ISO 3308:2000). Atmosphere for conditioning and testing are set out in ISO Standard 3402 (ISO 3402:1999).

[0321] The resistance to draw (RTD) of the downstream section may be at least 0 millimetres H.sub.2O. The RTD of the downstream section may be at least 3 millimetres H.sub.2O. The RTD of the downstream section may be at least 6 millimetres H.sub.2O.

[0322] The RTD of the downstream section may be no greater than 12 millimetres H.sub.2O. The RTD of the downstream section may be no greater than 11 millimetres H.sub.2O. The RTD of the downstream section may be no greater than 10 millimetres H.sub.2O.

[0323] The length of the downstream filter segment may be at least 5 millimetres. The length of the downstream filter segment may be at least 10 millimetres. The length of the downstream filter segment may be less than 25 millimetres. The length of the downstream filter segment may be less than 20 millimetres. For example, the length of the downstream filter segment may be between 5 millimetres and 25 millimetres, or between 10 millimetres and 25 millimetres, or between 5 millimetres and 20 millimetres, or between 10 millimetres and 20 millimetres.

[0324] The downstream section may further comprise one or more additional hollow tubular elements.

[0325] In certain embodiments, the downstream section may comprise a hollow tubular support element upstream of the hollow tubular cooling element described above. Preferably, the hollow tubular support element abuts the downstream end of the second aerosol-generating segment, or where present the third aerosol-generating segment. Preferably, the hollow tubular support element abuts the upstream end of the hollow tubular cooling element. Preferably, the hollow tubular support element and the hollow tubular cooling element are adjacent to each other and together provide a hollow tubular section within the downstream section.

[0326] Alternatively or in addition to the hollow tubular support element, the downstream section may further comprise a downstream hollow tubular element downstream of the hollow tubular cooling element. The downstream hollow tubular element may be provided immediately adjacent to the hollow tubular cooling element. Where the downstream section further comprises an additional downstream hollow tubular element, as described above, the additional downstream hollow tubular element may be formed of the same material as the downstream hollow tubular element, or a different material.

[0327] In certain preferred embodiments, the downstream section may comprise a ventilation zone

at a location on the downstream hollow tubular element. In one example, this ventilation zone at a location on the downstream hollow tubular element may be provided instead of a ventilation zone at a location on the hollow tubular cooling element. In another example, the ventilation zone at a location on the downstream hollow tubular element may be provided in addition to the ventilation zone provided at a location on the hollow tubular cooling element.

[0328] The ventilation zone at a location along the downstream hollow tubular element may comprise a plurality of perforations through the peripheral wall of the downstream hollow tubular element. Preferably, the ventilation zone at a location along the downstream hollow tubular element comprises at least one circumferential row of perforations. In some embodiments, the ventilation zone may comprise two circumferential rows of perforations. For example, the perforations may be formed online during manufacturing of the aerosol-generating article. Preferably, each circumferential row of perforations comprises from 8 to 30 perforations.

[0329] The downstream section may optionally further comprise may further comprise an additional cooling element defining a plurality of longitudinally extending channels such as to make a high surface area available for heat exchange. In other words, one such additional cooling element is adapted to function substantially as a heat exchanger. The plurality of longitudinally extending channels may be defined by a sheet material that has been pleated, gathered or folded to form the channels. The plurality of longitudinally extending channels may be defined by a single sheet that has been pleated, gathered or folded to form multiple channels. The sheet may also have been crimped prior to being pleated, gathered or folded. Alternatively, the plurality of longitudinally extending channels may be defined by multiple sheets that have been crimped, pleated, gathered or folded to form multiple channels. In some embodiments, the plurality of longitudinally extending channels may be defined by multiple sheets that have been crimped, pleated, gathered or folded together—that is by two or more sheets that have been brought into overlying arrangement and then crimped, pleated, gathered or folded as one.

[0330] As used herein, the term ‘crimped’ denotes a sheet having a plurality of substantially parallel ridges or corrugations. Preferably, when the aerosol-generating article has been assembled, the substantially parallel ridges or corrugations extend in a longitudinal direction. As used herein, the terms ‘gathered’, ‘pleated’, or ‘folded’ denote that a sheet of material is convoluted, folded, or otherwise compressed or constricted substantially transversely to the cylindrical axis of the aerosol-generating article. A sheet may be crimped prior to being gathered, pleated or folded. A sheet may be gathered, pleated or folded without prior crimping.

[0331] One such additional cooling element may have a total surface area of between about 300 square millimetre per millimetre length and about 1000 square millimetres per millimetre length.

[0332] The additional cooling element preferably offers a low resistance to the passage of air through additional cooling element. Preferably, the additional cooling element does not substantially affect the resistance to draw of the aerosol-generating article. To achieve this, it is preferred that the porosity in a longitudinal direction is greater than 50 percent and that the airflow path through the additional cooling element is relatively uninhibited. The longitudinal porosity of the additional cooling element may be defined by a ratio of the cross-sectional area of material forming the additional cooling element and an internal cross-sectional area of the aerosol-generating article at the portion containing the additional cooling element.

[0333] The additional cooling element preferably comprises a sheet material selected from the group comprising a metallic foil, a polymeric sheet, and a substantially non-porous paper or cardboard. In some embodiments, the aerosol-cooling element may comprise a sheet material selected from the group consisting of polyethylene (PE), polypropylene (PP), polyvinylchloride (PVC), polyethylene terephthalate (PET), polylactic acid (PLA), cellulose acetate (CA), and aluminium foil. In a particularly preferred embodiment, the additional cooling element comprises a sheet of PLA.

[0334] Preferably, an overall length of an aerosol-generating article in accordance with the

invention is at least 40 millimetres. More preferably, an overall length of an aerosol-generating article in accordance with the invention is at least 50 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is at least 60 millimetres. [0335] An overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 90 millimetres. More preferably, an overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 85 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 80 millimetres.

[0336] In some embodiments, an overall length of the aerosol-generating article is preferably from 40 millimetres to 70 millimetres, more preferably from 45 millimetres to 70 millimetres. In other embodiments, an overall length of the aerosol-generating article is preferably from 40 millimetres to 60 millimetres, more preferably from about 45 millimetres to about 60 millimetres. In further embodiments, an overall length of the aerosol-generating article is preferably from 40 millimetres to 50 millimetres, more preferably from 45 millimetres to 50 millimetres. In an exemplary embodiment, an overall length of the aerosol-generating article is about 45 millimetres.

[0337] In alternative embodiments, an overall length of the aerosol-generating article is preferably from 50 millimetres to 90 millimetres, more preferably from 60 millimetres to 90 millimetres, even more preferably from 70 millimetres to 90 millimetres. In other embodiments, an overall length of the aerosol-generating article is preferably from 50 millimetres to 85 millimetres, more preferably from 60 millimetres to 85 millimetres, even more preferably from 70 millimetres to 85 millimetres. In further embodiments, an overall length of the aerosol-generating article is preferably from 50 millimetres to 80 millimetres, more preferably from 60 millimetres to 80 millimetres, even more preferably from 70 millimetres to 80 millimetres. In an exemplary embodiment, an overall length of the aerosol-generating article is 75 millimetres.

[0338] The aerosol-generating article has an external diameter of at least 5 millimetres. Preferably, the aerosol-generating article has an external diameter of at least 6 millimetres. More preferably, the aerosol-generating article has an external diameter of at least 7 millimetres.

[0339] Preferably, the aerosol-generating article has an external diameter of less than or equal to about 12 millimetres. More preferably, the aerosol-generating article has an external diameter of less than or equal to about 10 millimetres. Even more preferably, the aerosol-generating article has an external diameter of less than or equal to about 8 millimetres.

[0340] In some embodiments, the aerosol-generating article has an external diameter from about 5 millimetres to about 12 millimetres, preferably from about 6 millimetres to about 12 millimetres, more preferably from about 7 millimetres to about 12 millimetres. In other embodiments, the aerosol-generating article has an external diameter from about 5 millimetres to about 10 millimetres, preferably from about 6 millimetres to about 10 millimetres, more preferably from about 7 millimetres to about 10 millimetres. In further embodiments, the aerosol-generating article has an external diameter from about 5 millimetres to about 8 millimetres, preferably from about 6 millimetres to about 8 millimetres, more preferably from about 7 millimetres to about 8 millimetres. In other embodiments, the aerosol-generating article has an external diameter of less than 7 millimetres.

[0341] The external diameter of the aerosol-generating article may be substantially constant over the whole length of the article. As an alternative, different portions of the aerosol-generating article may have different external diameters.

[0342] In particularly preferred embodiments, one or more of the components of the aerosol-generating article are individually circumscribed by their own wrapper.

[0343] In an embodiment, the first aerosol-generating segment, the second aerosol generating segment, and the mouthpiece element are individually wrapped. The upstream element (where present), the first aerosol-generating segment, the second aerosol generating segment, and the hollow tubular element are then combined together with an outer wrapper. Subsequently, they are

combined with the downstream filter element—which has its own wrapper— by means of tipping paper.

[0344] Preferably, at least one of the components of the aerosol-generating article is wrapped in a hydrophobic wrapper.

[0345] The term “hydrophobic” refers to a surface exhibiting water repelling properties. One useful way to determine this is to measure the water contact angle. The “water contact angle” is the angle, conventionally measured through the liquid, where a liquid/vapour interface meets a solid surface. It quantifies the wettability of a solid surface by a liquid via the Young equation. Hydrophobicity or water contact angle may be determined by utilizing TAPPI T558 test method and the result is presented as an interfacial contact angle and reported in “degrees” and can range from near zero to near 180 degrees.

[0346] In preferred embodiments, the hydrophobic wrapper is one including a paper layer having a water contact angle of about 30 degrees or greater, and preferably about 35 degrees or greater, or about 40 degrees or greater, or about 45 degrees or greater.

[0347] By way of example, the paper layer may comprise PVOH (polyvinyl alcohol) or silicon. The PVOH may be applied to the paper layer as a surface coating, or the paper layer may comprise a surface treatment comprising PVOH or silicon.

[0348] In certain embodiments of the invention, the aerosol-generating article further comprises one or more elongate susceptor elements within the one or both of the first aerosol-generating segment and the second aerosol generating segment. For example, one or more elongate susceptor elements may be arranged substantially longitudinally within one or both of the first aerosol-generating segment and the second aerosol generating segment and in thermal contact with one or both of the first and second aerosol-generating substrates.

[0349] As used herein with reference to the present invention, the term “susceptor element” refers to a material that can convert electromagnetic energy into heat. When located within a fluctuating electromagnetic field, eddy currents induced in the susceptor element cause heating of the susceptor element. As the susceptor element is located in thermal contact with the aerosol-generating substrate, the aerosol-generating substrate is heated by the susceptor element.

[0350] When used for describing the susceptor element, the term “elongate” means that the susceptor element has a length dimension that is greater than its width dimension or its thickness dimension, for example greater than twice its width dimension or its thickness dimension.

[0351] The susceptor element is arranged substantially longitudinally within at least one of the first and second aerosol-generating segments. This means that the length dimension of the elongate susceptor element is arranged to be approximately parallel to the longitudinal direction of at least one of the first and second aerosol-generating segments, for example within plus or minus 10 degrees of parallel to the longitudinal direction of at least one of the first and second aerosol-generating segments. In preferred embodiments, the elongate susceptor element may be positioned in a radially central position within the at least one of the first and second aerosol-generating segments, and extends along the longitudinal axis of at least one of the first and second aerosol-generating segments.

[0352] Preferably, the susceptor element extends all the way to a downstream end of at least one of the first and second aerosol-generating segments. In some embodiments, the susceptor element may extend all the way to an upstream end of at least one of the first and second aerosol-generating segments. In particularly preferred embodiments, the susceptor element has substantially the same length as the aerosol-generating segment within which it is included, and extends from the upstream end of the segment to the downstream end of the segment.

[0353] The susceptor element is preferably in the form of a pin, rod, strip or blade.

[0354] The susceptor element preferably has a length from 10 millimetres to 40 millimetres, for example from 15 millimetres to 35 millimetres, or from 17 millimetres to 30 millimetres.

[0355] The susceptor element preferably has a length from 5 millimetres to 15 millimetres, for

example from 6 millimetres to 12 millimetres, or from 8 millimetres to 10 millimetres.

[0356] The susceptor element preferably has a width from 1 millimetre to 5 millimetres.

[0357] The susceptor element may generally have a thickness from 0.01 millimetres to 2 millimetres, for example from 0.5 millimetres to 2 millimetres. In some embodiments, the susceptor element preferably has a thickness from 10 micrometres to 500 micrometres, more preferably from 10 micrometres to 100 micrometres.

[0358] Preferably, the elongate susceptor element has a length which is the same or shorter than the length of the aerosol-generating segment in which it is incorporated. Preferably, the elongate susceptor element has a same length as the aerosol-generating segment in which it is incorporated.

[0359] The susceptor element may be formed from any material that can be inductively heated to a temperature sufficient to generate an aerosol from the aerosol-generating substrate. Preferred susceptor elements comprise a metal or carbon.

[0360] A preferred susceptor element may comprise or consist of a ferromagnetic material, for example a ferromagnetic alloy, ferritic iron, or a ferromagnetic steel or stainless steel. A suitable susceptor element may be, or comprise, aluminium. Preferred susceptor elements may be formed from 400 series stainless steels, for example grade 410, or grade 420, or grade 430 stainless steel. Different materials will dissipate different amounts of energy when positioned within electromagnetic fields having similar values of frequency and field strength.

[0361] Thus, parameters of the susceptor element such as material type, length, width, and thickness may all be altered to provide a desired power dissipation within a known electromagnetic field. Preferred susceptor elements may be heated to a temperature in excess of 250 degrees Celsius.

[0362] Suitable susceptor elements may comprise a non-metallic core with a metal layer disposed on the non-metallic core, for example metallic tracks formed on a surface of a ceramic core. A susceptor element may have a protective external layer, for example a protective ceramic layer or protective glass layer encapsulating the susceptor element. The susceptor element may comprise a protective coating formed by a glass, a ceramic, or an inert metal, formed over a core of susceptor element material.

[0363] The susceptor element is arranged in thermal contact with the aerosol-generating substrate of the aerosol-generating segment in which the susceptor element is incorporated. Thus, when the susceptor element heats up the aerosol-generating substrate is heated up and an aerosol is formed. Preferably the susceptor element is arranged in direct physical contact with the aerosol-generating substrate, for example within the aerosol-generating substrate.

[0364] The first aerosol-generating substrate may be circumscribed by a first wrapper. The second aerosol-generating substrate may be circumscribed by a second wrapper, separate from the first wrapper. Each aerosol-generating segment therefore has its own distinct plug wrapper around the respective aerosol-generating substrate.

[0365] The first aerosol-generating segment and the second aerosol-generating segment may be circumscribed by a further wrapper, which may combine the aerosol-generating segments and retain them in position relative to each other.

[0366] The wrapper circumscribing the first aerosol-generating substrate, the second aerosol-generating substrate, or the first aerosol-generating segment and the second aerosol-generating segment may be a paper wrapper or a non-paper wrapper. Suitable paper wrappers for use in specific embodiments of the invention are known in the art and include, but are not limited to: cigarette papers; and filter plug wraps. Suitable non-paper wrappers for use in specific embodiments of the invention are known in the art and include, but are not limited to sheets of homogenised tobacco materials.

[0367] A paper wrapper may have a grammage of at least 15 gsm (grams per square metre), preferably at least 20 gsm. The paper wrapper may have a grammage of less than or equal to 35 gsm, preferably less than or equal to 30 gsm. The paper wrapper may have a grammage from 15

gsm to 35 gsm, preferably from 20 gsm to 30 gsm. In a preferred embodiment, the paper wrapper may have a grammage of 25 gsm. A paper wrapper may have a thickness of at least 25 micrometres, preferably at least 30 micrometres, more preferably at least 35 micrometres. The paper wrapper may have a thickness of less than or equal to 55 micrometres, preferably less than or equal to 50 micrometres, more preferably less than or equal to 45 micrometres. The paper wrapper may have a thickness from 25 micrometres to 55 micrometres, preferably from 30 micrometres to 50 micrometres, more preferably from 35 micrometres to 45 micrometres. In a preferred embodiment, the paper wrapper may have a thickness of 40 microns.

[0368] In certain preferred embodiments, the wrapper may be formed of a laminate material comprising a plurality of layers. Preferably, the wrapper is formed of an aluminium co-laminated sheet. The use of a co-laminated sheet comprising aluminium advantageously prevents combustion of the aerosol-generating substrate in the event that the aerosol-generating substrate should be ignited, rather than heated in the intended manner.

[0369] A paper layer of the co-laminated sheet may have a grammage of at least 35 gsm, preferably at least 40 gsm. The paper layer of the co-laminated sheet may have a grammage of less than or equal to 55 gsm, preferably less than or equal to 50 gsm. The paper layer of the co-laminated sheet may have a grammage from 35 gsm to 55 gsm, preferably from 40 gsm to 50 gsm. In a preferred embodiment, the paper layer of the co-laminated sheet may have a grammage of 45 gsm.

[0370] A paper layer of the co-laminated sheet may have a thickness of at least 50 micrometres, preferably at least 55 micrometres, more preferably at least 60 micrometres. The paper layer of the co-laminated sheet may have a thickness of less than or equal to 80 micrometres, preferably less than or equal to 75 micrometres, more preferably less than or equal to 70 micrometres.

[0371] The paper layer of the co-laminated sheet may have a thickness from 50 micrometres to 80 micrometres, preferably from 55 micrometres to 75 micrometres, more preferably from 60 micrometres to 70 micrometres. In a preferred embodiment, the paper layer of the co-laminated sheet may have a thickness of 65 microns.

[0372] A metallic layer of the co-laminated sheet may have a grammage of at least 12 gsm, preferably at least 15 gsm. The metallic layer of the co-laminated sheet may have a grammage of less than or equal to 25 gsm, preferably less than or equal to 20 gsm. The metallic layer of the co-laminated sheet may have a grammage from 12 gsm to 25 gsm, preferably from 15 gsm to 20 gsm. In a preferred embodiment, the metallic layer of the co-laminated sheet may have a grammage of 17 gsm.

[0373] A metallic layer of the co-laminated sheet may have a thickness of at least 2 micrometres, preferably at least 3 micrometres, more preferably at least 5 micrometres. The metallic layer of the co-laminated sheet may have a thickness of less than or equal to 15 micrometres, preferably less than or equal to 12 micrometres, more preferably less than or equal to 10 micrometres.

[0374] The metallic layer of the co-laminated sheet may have a thickness from 2 micrometres to 15 micrometres, preferably from 3 micrometres to 12 micrometres, more preferably from 5 micrometres to 10 micrometres. In a preferred embodiment, the metallic layer of the co-laminated sheet may have a thickness of 6 microns.

[0375] The wrapper circumscribing the first aerosol-generating segment and the second aerosol-generating segment may be a paper wrapper comprising PVOH (polyvinyl alcohol) or silicone (or polysiloxane) (or polysiloxane). Addition of PVOH (polyvinyl alcohol) or silicone (or polysiloxane) may improve the grease barrier properties of the wrapper.

[0376] The PVOH or silicone (or polysiloxane) may be applied to the paper layer as a surface coating, such as disposed on an exterior surface of the paper layer of the wrapper circumscribing the first aerosol-generating segment and the second aerosol-generating segment. The PVOH or silicone (or polysiloxane) may be disposed on and form a layer on the exterior surface of the paper layer of the wrapper. The PVOH or silicone (or polysiloxane) may be disposed on an interior surface of the paper layer of the wrapper. The PVOH or silicone (or polysiloxane) may be disposed

on and form a layer on the interior surface of the paper layer of the aerosol generating article.

[0377] The paper wrapper comprising PVOH or silicone (or polysiloxane) may have a grammage of at least 20 gsm, preferably at least 25 gsm, more preferably at least 30 gsm. The paper wrapper comprising PVOH or silicone (or polysiloxane) may have a grammage of less than or equal to 50 gsm, preferably less than or equal to 45 gsm, more preferably less than or equal to 40 gsm.

[0378] The paper wrapper comprising PVOH or silicone (or polysiloxane) may have a thickness of at least 25 micrometres, preferably at least 30 micrometres, more preferably at least 35 micrometres. The paper wrapper comprising PVOH or silicone (or polysiloxane) may have a thickness of less than or equal to 50 micrometres, preferably less than or equal to 45 micrometres, more preferably less than or equal to 40 micrometres.

[0379] The wrapper circumscribing the first aerosol-generating segment and the second aerosol-generating segment may comprise a flame retardant composition comprising one or more flame retardant compounds. The term “flame retardant compounds” is used herein to describe chemical compounds that, when added to or otherwise incorporated into a carrier substrate, such as paper or plastic compounds, provide the carrier substrate with varying degrees of flammability protection. In practice, flame retardant compounds may be activated by the presence of an ignition source and are adapted to prevent or slow the further development of ignition by a variety of different physical and chemical mechanisms.

[0380] A flame retardant composition may typically further comprise one or more non-flame retardant compounds, that is, one or more compound—such as a solvent, an excipient, a filler—that does not actively contribute to providing the carrier substrate with flammability protection, but is used to facilitate the application of the flame retardant compound or compounds onto or into the wrapper or both. Some of the non-flame retardant compounds of a flame retardant composition—such as solvents—are volatile and may evaporate from the wrapper upon drying after the flame retardant composition has been applied onto or into the wrapping base material or both. As such, although such non-flame retardant compounds form part of the formulation of the flame retardant composition, they may no longer be present or they may only be detectable in trace amounts in the wrapper of an aerosol-generating article.

[0381] For example, the flame retardant composition may comprise a polymer and a mixed salt based on at least one mono, di- and/or tri-carboxylic acid, at least one polyphosphoric, pyrophosphoric and/or phosphoric acid, and a hydroxide or a salt of an alkali or an alkaline earth metal, where the at least one mono, di- and/or tri-carboxylic acid and the hydroxide or salt form a carboxylate and the at least one polyphosphoric, pyrophosphoric and/or phosphoric acid and the hydroxide or salt form a phosphate. Preferably, the flame retardant composition may further comprise a carbonate of an alkali or an alkaline earth metal.

[0382] Below, there is provided a non-exhaustive list of non-limiting examples. Any one or more of the features of these examples may be combined with any one or more features of another example, embodiment, or aspect described herein.

[0383] Example 1: An aerosol-generating system comprising; an aerosol-generating device for use with an aerosol-generating article, the aerosol-generating device comprising: a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone, and an aerosol-generating article for being received in the heating chamber of the aerosol-generating device, the aerosol-generating article comprising: a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate, the aerosol-generating system being configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device: no more than 50 percent of the length of the first aerosol-generating segment is within the heating zone, and at least 90 percent of the length of the second aerosol-generating segment is within the heating zone.

[0384] Example 1a: An aerosol-generating system comprising; an aerosol-generating device for use

with an aerosol-generating article, the aerosol-generating device comprising: a heating chamber for receiving an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone, and an aerosol-generating article for being received in the heating chamber of the aerosol-generating device, the aerosol-generating article comprising: a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate, the aerosol-generating system being configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device: at least 90 percent of the length of the second aerosol-generating segment is within the heating zone.

[0385] Example 1b: An aerosol-generating system according to Example 1 or Example 1a, wherein the first aerosol-generating segment is located upstream of the second aerosol-generating segment.

[0386] Example 1c: An aerosol-generating system according to any preceding Example, the length of the first aerosol-generating segment is less than the length of the second aerosol-generating segment.

[0387] Example 1d: An aerosol-generating system according to any preceding Example, wherein the second aerosol-generating segment has a length of between 9 millimetres and 15 millimetres.

[0388] Example 1e: An aerosol-generating system according to any preceding Example, wherein the combined length of the first aerosol-generating segment and the second aerosol-generating segment is between 15 millimetres and 25 millimetres.

[0389] Example 3: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating segment has an upstream end, the upstream end of the first aerosol-generating segment defining the upstream end of the aerosol-generating article.

[0390] Example 4: An aerosol-generating system according to any preceding Example, wherein the upstream end of the second aerosol-generating segment is in direct contact with the downstream end of the first aerosol-generating segment.

[0391] Example 6: An aerosol-generating system according to any preceding Example, wherein the length of the second aerosol-generating segment is at least 3 millimetres greater than the length of the first aerosol-generating segment.

[0392] Example 8: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating segment has a length of at least 2 millimetres.

[0393] Example 9: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating segment has a length of no more than 8 millimetres.

[0394] Example 10: An aerosol-generating system according to any preceding Example, wherein the second aerosol-generating segment has a length of at least 8 millimetres.

[0395] Example 11: An aerosol-generating system according to any preceding Example, wherein the second aerosol-generating segment has a length of no more than 16 millimetres.

[0396] Example 12: An aerosol-generating system according to any preceding Example, wherein the heater assembly comprises at least one of a resistive heating element and an inductive heating assembly.

[0397] Example 12a: An aerosol-generating system according to any preceding Example, wherein the heater assembly comprises an inductively heated element and an inductor coil.

[0398] Example 13: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating substrate has a first density and the second aerosol-generating substrate has a second density, the second density being greater than the first density.

[0399] Example 14. An aerosol-generating article according to any preceding Example, wherein the density of the second aerosol-generating substrate is at least 100 mg per cubic centimetre higher than the density of the first aerosol-generating substrate.

[0400] Example 15: An aerosol-generating system according to any preceding Example, wherein the density of the first aerosol-generating substrate is less than 400 mg per cubic centimetre.

[0401] Example 16. An aerosol-generating article according to any preceding Example, wherein the

first aerosol-generating substrate has a density of between 100 mg per cubic centimetre and 400 mg per cubic centimetre.

[0402] Example 17: An aerosol-generating system according to any preceding Example, wherein the density of the second aerosol-generating substrate is at least 500 mg per cubic centimetre.

[0403] Example 18. An aerosol-generating article according to any preceding Example, wherein the second aerosol-generating substrate has a density of between 500 mg per cubic centimetre and 1000 mg per cubic centimetre.

[0404] Example 19. An aerosol-generating article according to any preceding Example, wherein the density of the second aerosol-generating substrate is at least 1.2 times the density of the first aerosol-generating substrate.

[0405] Example 20. An aerosol-generating article according to any preceding Example, wherein the first aerosol-generating substrate comprises shredded tobacco material.

[0406] Example 21: An aerosol-generating system according to Example 20, wherein the first aerosol-generating substrate comprises tobacco cut filler.

[0407] Example 22. An aerosol-generating article according to any preceding Example, wherein the first aerosol-generating substrate comprises a shredded sheet of homogenised tobacco material.

[0408] Example 23. An aerosol-generating article according to any preceding Example, wherein the second aerosol-generating substrate comprises an aerosol-generating film.

[0409] Example 24: An aerosol-generating system according to Example 23, wherein the second aerosol-generating substrate comprises an aerosol-generating film, the aerosol-generating film comprising a cellulosic based film forming agent, nicotine and glycerol, wherein the aerosol-generating film has a glycerol content of at least 40 percent by weight.

[0410] Example 25. An aerosol-generating article according to Example 24, wherein the second aerosol-generating substrate comprises at least 50 percent by weight of glycerol.

[0411] Example 26. An aerosol-generating article according to any one of Examples 23 to 25, wherein the aerosol-generating film comprises a cellulosic film forming agent, nicotine, and aerosol former.

[0412] Example 27. An aerosol-generating article according to any one of Examples 23 to 26, wherein the aerosol-generating film further comprises a cellulose based strengthening agent.

[0413] Example 28. An aerosol-generating article according to any one of Examples 23 to 27, wherein the aerosol-generating film further comprises a carboxymethyl cellulose.

[0414] Example 29. An aerosol-generating article according to any one of Examples 23 to 28, wherein the aerosol-generating film further comprises an acid.

[0415] Example 30. An aerosol-generating article according to any of Examples 23 to 29, wherein the aerosol-generating film is substantially tobacco free.

[0416] Example 31. An aerosol-generating article according to any preceding Example, wherein the second aerosol-generating substrate comprises a gel composition comprising nicotine, at least one gelling agent and, aerosol former.

[0417] Example 32: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating substrate and the second aerosol-generating substrate comprise at least one aerosol former, the aerosol former content of the second aerosol-generating substrate being greater than the aerosol former content of the first aerosol-generating substrate.

[0418] Example 33: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the first aerosol-generating substrate is no more than 30 percent by weight, on a dry weight basis.

[0419] Example 34: An aerosol-generating system according to Example 32 or Example 33, wherein the second aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight, on a dry weight basis.

[0420] Example 35. An aerosol-generating article according to any one of Examples 32 to 34, wherein the aerosol former content of the second aerosol-generating substrate is at least 15 percent higher than the aerosol former content of the first aerosol-generating substrate.

[0421] Example 36. An aerosol-generating article according to any one of Examples 32 to 35, wherein the aerosol former content of the second aerosol-generating substrate is at least 1.2 times the aerosol former content of the first aerosol-generating substrate.

[0422] Example 37: An aerosol-generating article according to any one of Examples 32 to 36, wherein the aerosol former content of the first aerosol-generating substrate is between 5 percent and 30 percent by weight on a dry weight basis.

[0423] Example 38. An aerosol-generating article according to any one of Examples 32 to 37, wherein the aerosol former content of the second aerosol-generating substrate is between 40 percent and 80 percent by weight on a dry weight basis.

[0424] Example 39: An aerosol-generating system according to any preceding Example, wherein the first aerosol-generating substrate of the first aerosol-generating segment is circumscribed by a first wrapper, and wherein the second aerosol-generating substrate of the second aerosol-generating segment is circumscribed by a second wrapper.

[0425] Example 40. An aerosol-generating article according to any preceding Example, wherein the aerosol-generating rod further comprises a third aerosol-generating segment provided upstream of the second aerosol-generating segment and comprising a third aerosol-generating substrate.

[0426] Example 41. An aerosol-generating article according to Example 40, wherein the third aerosol-generating substrate comprises shredded tobacco material.

[0427] Example 42. An aerosol-generating article according to Example 40 or Example 41, wherein the third aerosol-generating substrate comprises at least one aerosol former and wherein the aerosol former content of the third aerosol-generating substrate is no more than 30 percent by weight on a dry weight basis.

[0428] Example 43. An aerosol-generating article according to any preceding Example, further comprising a downstream section provided downstream of the first aerosol-generating segment and the second aerosol-generating segment.

[0429] Example 44. An aerosol-generating article according to Example 43, wherein the downstream section extends to a downstream end of the aerosol-generating article.

[0430] Example 45. An aerosol-generating article according to Example 43 or Example 44, wherein the downstream section comprises a downstream filter segment.

[0431] Example 46. An aerosol-generating article according to Example 45, wherein the downstream filter segment is a solid plug.

[0432] Example 47. An aerosol-generating article according to Example 44 and Example 46, wherein the downstream filter segment has a length of at least 5 millimetres.

[0433] Example 48. An aerosol-generating article according to any one of Examples 43 to Example 47, wherein the downstream section comprises a hollow tubular cooling element.

[0434] Example 49. An aerosol-generating article according to Example 48, wherein the hollow tubular cooling element has a length of at least 20 millimetres.

[0435] Example 50. An aerosol-generating article according to Example 48 or Example 49, wherein the downstream section comprises a ventilation zone at a location along the hollow tubular cooling element.

[0436] Example 51. An aerosol-generating article according to any one of Examples 48 to 50, wherein the downstream section further comprises a hollow tubular support element upstream of the hollow tubular cooling element.

[0437] Example 52. An aerosol-generating article according to any one of Examples 48 to 51, wherein the downstream section further comprises a downstream hollow tubular element downstream of the hollow tubular cooling element.

[0438] Example 53. An aerosol-generating article according to any preceding example, further

comprising an upstream element provided upstream of the first aerosol-generating segment and the second aerosol-generating segment.

[0439] Example 54. An aerosol-generating article according to any preceding example, wherein the aerosol-generating article has a ventilation level of at least 40 percent.

[0440] Example 55. An aerosol-generating article according to any preceding example, wherein the length of the aerosol-generating article is between 40 millimetres and 50 millimetres.

[0441] Example 56. An aerosol-generating article according to any one of Examples 1 to 55, wherein the length of the aerosol-generating article is between 70 millimetres and 80 millimetres.

Description

[0442] In the following, the invention will be further described with reference to the drawings of the accompanying Figures, in which:

[0443] FIG. 1 shows a schematic side perspective view of a first aerosol-generating article for use in an aerosol-generating system according to the present invention;

[0444] FIG. 2 shows a schematic side sectional view of a first aerosol-generating system according to the present invention, the first aerosol-generating system comprising the first aerosol-generating article of FIG. 1;

[0445] FIG. 3 shows a schematic side sectional view of a second aerosol-generating system according to the present invention, the second aerosol-generating system comprising a second aerosol-generating article;

[0446] FIG. 4 shows a schematic side perspective view of a third aerosol-generating article for use in an aerosol-generating system according to the present invention;

[0447] FIG. 5 shows a schematic side sectional view of a third aerosol-generating system according to the present invention, the third aerosol-generating system comprising the third aerosol-generating article of FIG. 4;

[0448] The aerosol-generating article **10** shown in FIG. 1 comprises a first aerosol-generating segment **24**, a second aerosol-generating segment **26**, and a downstream section **14**. The upstream end of the first aerosol-generating segment **24** defines the upstream end **16** of the aerosol-generating article **10**. The second aerosol-generating segment **26** is located immediately downstream of the first aerosol-generating segment **24**. The downstream end of the first aerosol-generating segment **24** abuts the upstream end of the second aerosol-generating segment **26**. The downstream section **14** is located immediately downstream of the second aerosol-generating segment **26**. The downstream end of the second aerosol-generating segment **26** abuts the upstream end of the downstream section **14**. The downstream end of the downstream section **14** defines the downstream end **18** of the aerosol-generating article **10**.

[0449] The downstream section **14** comprises a hollow tubular cooling element **20** and a downstream filter segment **50**.

[0450] The aerosol-generating article **10** has an overall length of about 45 millimetres and an external diameter of about 7.2 mm.

[0451] The first aerosol-generating segment **24** has a length of 5 millimetres and comprises a first aerosol-generating substrate formed of about 50 mg of shredded tobacco material comprising between 15 percent by weight and 20 percent by weight of glycerol. The density of the first aerosol-generating substrate is about 300 mg per cubic centimetre. The first aerosol-generating segment **24** is individually wrapped by a plug wrap (not shown).

[0452] The second aerosol-generating segment **26** has a length of 12 millimetres and comprises a second aerosol-generating substrate formed of shreds of an aerosol-generating film. Example compositions for the aerosol-generating film are shown below in Table 1:

TABLE-US-00001 TABLE 1 aerosol-generating film compositions Composition A Composition B

Component Amount (% by weight) Amount (% by weight) HPMC 22.39 23.71 Cellulose fibres 16.79 0 Cellulose powder 0 17.79 Sodium 5.60 5.93 carboxymethylcellulose Nicotine 1.40 1.48 Glycerol 50.37 47.43 Lactic acid 3.45 3.66

[0453] The second aerosol-generating substrate has a glycerol content of around 50 percent by weight, as shown above, which is therefore over 10 percent higher than the glycerol content of the first aerosol-generating substrate. The density of the second aerosol-generating substrate is above 600 mg per cubic centimetre. The first aerosol-generating segment **24** is individually wrapped by a plug wrap (not shown).

[0454] The hollow tubular cooling element **20** of the downstream section **14** is located immediately downstream of the aerosol-generating rod **12**, the hollow tubular cooling element **20** being in longitudinal alignment with the rod **12**. The upstream end of the hollow tubular cooling element **20** abuts the downstream end of the rod **12**.

[0455] The hollow tubular cooling element **20** defines a hollow section of the aerosol-generating article **10**. The hollow tubular cooling element **20** does not substantially contribute to the overall RTD of the aerosol-generating article. In more detail, an RTD of the hollow tubular cooling element **20** is about 0 mm H.sub.2O.

[0456] The hollow tubular cooling element **20** is provided in the form of a hollow cylindrical tube made of cardboard. The hollow tubular cooling element **20** defines an internal cavity that extends all the way from an upstream end of the hollow tubular cooling element **20** to a downstream end of the hollow tubular cooling element **20**. The internal cavity is substantially empty, and so substantially unrestricted airflow is enabled along the internal cavity.

[0457] The hollow tubular cooling element **20** has a length of about 21 millimetres, an external diameter of about 7.2 millimetres, and an internal diameter of about 6.7 millimetres. Thus, a thickness of a peripheral wall of the hollow tubular cooling element **20** is about 0.25 millimetres.

[0458] The aerosol-generating article **10** comprises a ventilation zone **30** provided at a location along the hollow tubular cooling element **20**. The ventilation zone **30** comprises a circumferential row of openings or perforations circumscribing the hollow tubular cooling element **20**. The perforations of the ventilation zone **30** extend through the wall of the hollow tubular cooling element **20**, in order to allow fluid ingress into the internal cavity from the exterior of the article **10**. A ventilation level of the aerosol-generating article **10** is about 40 percent.

[0459] The downstream filter segment **50** extends from the downstream end of the hollow tubular cooling element **20** to the downstream or mouth end of the aerosol-generating article **10**. The downstream filter segment **50** has a length of about 7 millimetres. An external diameter of the downstream filter segment **50** is about 7.2 millimetres. The downstream filter segment **50** comprises a low-density, cellulose acetate filter segment. The RTD of the downstream filter segment **50** is about 8 mm H.sub.2O. The downstream filter segment **50** may be individually wrapped by a plug wrap (not shown).

[0460] The article **10** comprises an upstream wrapper **44** circumscribing the first aerosol-generating segment **24**, the second aerosol-generating segment **26**, and the hollow tubular cooling element **20**. The ventilation zone **30** may also comprise a circumferential row of perforations provided on the upstream wrapper **44**. The perforations of the upstream wrapper **44** overlap the perforations provided on the hollow tubular cooling element **20**. Accordingly, the upstream wrapper **44** overlies the perforations of the ventilation zone **30** provided on the hollow tubular cooling element **20**.

[0461] The article **10** also comprises a tipping wrapper **52** circumscribing the hollow tubular cooling element **20** and the mouthpiece element **50**. The tipping wrapper **52** overlies the portion of the upstream wrapper **44** that overlies the hollow tubular cooling element **20**. This way the tipping wrapper **52** effectively joins the mouthpiece element **50** to the rest of the components of the article **10**. The width of the tipper wrapper **52** is about 26 millimetres. Additionally, the ventilation zone **30** may comprise a circumferential row of perforations provided on the tipping wrapper **52**. The perforations of the tipping wrapper **52** overlap the perforations provided on the hollow tubular

cooling element **20** and the upstream wrapper **44**. Accordingly, the tipping wrapper **52** overlies the perforations of the ventilation zone **30** provided on the hollow tubular cooling element **20** and the upstream wrapper **44**.

[0462] FIG. 2 illustrates a first aerosol-generating system **100** according to the present invention. The first aerosol-generating system **100** comprises the first aerosol-generating article **10** of FIG. 1, and a downstream portion of an aerosol-generating device **1**. The aerosol-generating device **1** comprises a housing (or body) **4**, extending between a downstream end **2** and an upstream end (not shown). The housing **4** comprises a peripheral wall **6**. The peripheral wall **6** defines a heating chamber **3** for receiving an aerosol-generating article **10**. The heating chamber **3** is defined by a closed, upstream end and an open, downstream end. The downstream end of the heating chamber **3** is located at the downstream end of the aerosol-generating device **1**. The aerosol-generating article **10** is configured to be received through the open, downstream end of the heating chamber **3** and is configured to abut a closed, upstream end of the device cavity, when the aerosol-generating article **10** is fully received in the heating chamber **3**.

[0463] A device air flow channel **5** is defined within the peripheral wall **6**. The air-flow channel **5** extends between an inlet **7** located at the mouth end of the aerosol-generating device **1** and the closed end of the device cavity. Air may enter the aerosol-generating substrate **12** via an aperture (not shown) provided at the closed end of the device cavity, ensuring fluid communication between the air flow channel **5** and the aerosol-generating substrate **12**.

[0464] The aerosol-generating device **1** further comprises a heater assembly **15** and a power source (not shown) for supplying power to the heater assembly **15**. A controller (not shown) is also provided to control such supply of power to the heater assembly **15**. The heater assembly **15** is configured to controllably heat the aerosol-generating article **10** during use, when the aerosol-generating article **1** is fully received within the heating chamber **3**.

[0465] The heater assembly **15** extends from an upstream end to a downstream end defining a heating zone. The heater assembly **15** is the same length as the second aerosol-generating segment **26** such that when the aerosol-generating article **10** is fully received within the heating chamber **3**, the entire length of the second aerosol-generating segment **26** is received within the heating zone to provide optimal heating of the aerosol-generating segment **26**. When the aerosol-generating article **10** is fully received within the heating chamber **3**, the first aerosol-generating segment **24** is disposed upstream of the heating zone such that none of the length of the first aerosol-generating segment **24** is disposed within the heating zone. The heater assembly **15** comprises a resistive heating element. When the aerosol-generating article **10** is fully received within the heating chamber **3**, an upstream portion of the hollow tubular cooling element **20** is also received within the heating chamber **3**. Such an upstream portion of the hollow tubular cooling element **20** is 11 millimetres in length. Accordingly, about 28 millimetres of the article **10** is received within the heating chamber **3** and about 17 millimetres of the article **10** is located outside of the heating chamber **3**. In other words, about 17 millimetres of the article **10** protrudes from the device **1** when the article **10** is fully received within the heating chamber **3**. Such a length, PL, of the article **10** protruding from the device **1** is shown in FIG. 2.

[0466] The ventilation zone **30** is arranged to be exposed when the aerosol-generating article **10** is fully received within the heating chamber **3**.

[0467] In use, the aerosol-generating article **10** is fully received within the heating chamber **3** of the aerosol-generating device **1**. The heater assembly **15** is activated by the controller and the resistive heating element generates heat which is transferred directly to the second aerosol-generating segment **26** which is disposed within the heating zone. This generates an aerosol in the second aerosol-generating substrate. The first aerosol-generating segment **24** which is not located within the heating zone is heated more slowly and to a lower temperature than the second aerosol-generating segment **26**. Nevertheless, because of the density and aerosol former content, the first aerosol-generating substrate also generates an aerosol. When a pressure drop is applied to the

downstream end of the aerosol-generating article **10**, air is drawn into the air inlet **7** and along the air flow channel **5** and into the first aerosol-generating segment **24**. The aerosols generated in the first and second aerosol generating substrates is entrained in the airflow which then passes through the downstream section before leaving through the downstream end of the aerosol-generating article.

[0468] FIG. **3** shows a second aerosol-generating system **200** according to the present invention. The second aerosol-generating system **200** comprises an aerosol-generating device **1** which is the same as the aerosol-generating device **1** described above in relation to the FIG. **2** embodiment. The second aerosol-generating system **200** further comprises an aerosol-generating article **12**. The aerosol-generating article **12** includes all of the same features of the aerosol-generating article **10** of the FIGS. **1** and **2** embodiment.

[0469] The aerosol-generating article **12** of FIG. **3** differs from the aerosol-generating article **10** of FIGS. **1** and **2** since the first aerosol-generating segment **24** has a length of 8.5 millimetres, and the second aerosol-generating segment has a length of 8.5 millimetres. This means that when the aerosol-generating article **12** is fully received in the heating chamber **3** of the aerosol-generating article **1**, a downstream portion of the first aerosol-generating segment **24** is disposed within the heating zone of the heater assembly **15**. In particular, in the aerosol-generating system **200** of FIG. **3** the downstream 3.5 millimetres of the first aerosol-generating segment **24** is disposed within the heating zone of the heater assembly **15**. This means that in use, a portion of the first aerosol-generating segment **24** is directly heated by the heater assembly.

[0470] FIG. **4** shows a third aerosol-generating article for use in an aerosol-generating system according to the present invention. The aerosol-generating article **110** shown in FIG. **4** comprises all the same features as the FIG. **1** aerosol-generating article **10**. However, in the aerosol-generating article **110** of FIG. **4**, the second aerosol-generating segment **126** has a length of 7 millimetres. A third aerosol-generating segment **128** is provided between the second aerosol-generating segment **126** and the downstream section **14**. The third aerosol-generating segment **128** has a length of 5 millimetres.

[0471] The third aerosol-generating segment **128** comprises a third aerosol-generating substrate. The third aerosol-generating substrate comprises shredded tobacco material comprising between 15 percent by weight and 20 percent by weight of glycerol. The bulk density of the third aerosol-generating substrate is about 250 mg per cubic centimetre. The third aerosol-generating segment **128** is individually wrapped by a plug wrap (not shown).

[0472] FIG. **5** shows a third aerosol-generating system according to the present invention. The third aerosol-generating system **300** comprises an aerosol-generating device **1** which is the same as the aerosol-generating device **1** described above in relation to the FIG. **2** embodiment. The aerosol-generating system **300** further comprises the aerosol-generating article **110** of FIG. **4**.

[0473] As can be seen, when the aerosol-generating article **110** is fully received within the heating chamber **3** of the aerosol-generating device, both the second and third aerosol-generating segments are within the heating zone of the heater assembly **15**.

[0474] For the purpose of the present description and of the appended claims, except where otherwise indicated, all numbers expressing amounts, quantities, percentages, and so forth, are to be understood as being modified in all instances by the term “about”. Also, all ranges include the maximum and minimum points disclosed and include any intermediate ranges therein, which may or may not be specifically enumerated herein. In this context, therefore, a number A is understood as $A \pm 10\%$ of A. Within this context, a number A may be considered to include numerical values that are within general standard error for the measurement of the property that the number A modifies. The number A, in some instances as used in the appended claims, may deviate by the percentages enumerated above provided that the amount by which A deviates does not materially affect the basic and novel characteristic(s) of the claimed invention. Also, all ranges include the

maximum and minimum points disclosed and include any intermediate ranges therein, which may or may not be specifically enumerated herein.

Claims

1.-12. (canceled)

13. An aerosol-generating system, comprising: an aerosol-generating device for an aerosol-generating article, the aerosol-generating device comprising: a heating chamber configured to receive an aerosol-generating article, and a heater assembly arranged along a portion of the heating chamber defining a heating zone; and an aerosol-generating article configured to be received in the heating chamber of the aerosol-generating device, the aerosol-generating article comprising: a first aerosol-generating segment comprising a first aerosol-generating substrate, and a second aerosol-generating segment comprising a second aerosol-generating substrate, the first aerosol-generating substrate comprising between 5 percent and 30 percent by weight of glycerol on a dry weight basis, the aerosol-generating system being configured such that when the aerosol-generating article is fully received within the heating chamber of the aerosol-generating device: at least 90 percent of a length of the second aerosol-generating segment is within the heating zone, wherein the first aerosol-generating segment is located upstream of the second aerosol-generating segment, and wherein a length of the first aerosol-generating segment is less than the length of the second aerosol-generating segment.

14. The aerosol-generating system according to claim 13, wherein the first aerosol-generating segment has an upstream end, the upstream end of the first aerosol-generating segment defining the upstream end of the aerosol-generating article.

15. The aerosol-generating system according to claim 13, wherein the upstream end of the second aerosol-generating segment is in direct contact with the downstream end of the first aerosol-generating segment.

16. The aerosol-generating system according to claim 13, wherein the first aerosol-generating segment has a length of at least 2 millimetres.

17. The aerosol-generating system according to claim 13, wherein the first aerosol-generating segment has a length of no more than 8 millimetres.

18. The aerosol-generating system according to claim 13, wherein the second aerosol-generating segment has a length of at least 8 millimetres.

19. The aerosol-generating system according to claim 13, wherein the second aerosol-generating segment has a length of no more than 16 millimetres.

20. The aerosol-generating system according to claim 13, wherein the heater assembly comprises at least one of a resistive heating element and an inductive heating assembly.

21. The aerosol-generating system according to claim 13, wherein the first aerosol-generating substrate has a first density and the second aerosol-generating substrate has a second density, the second density being greater than the first density.

22. The aerosol-generating system according to claim 13, wherein the first aerosol-generating substrate and the second aerosol-generating substrate comprise at least one aerosol former, an aerosol former content of the second aerosol-generating substrate being greater than an aerosol former content of the first aerosol-generating substrate.

23. The aerosol-generating system according to claim 13, wherein the second aerosol-generating substrate further comprises at least one aerosol former, and wherein the aerosol former content of the second aerosol-generating substrate is at least 40 percent by weight on a dry weight basis.
